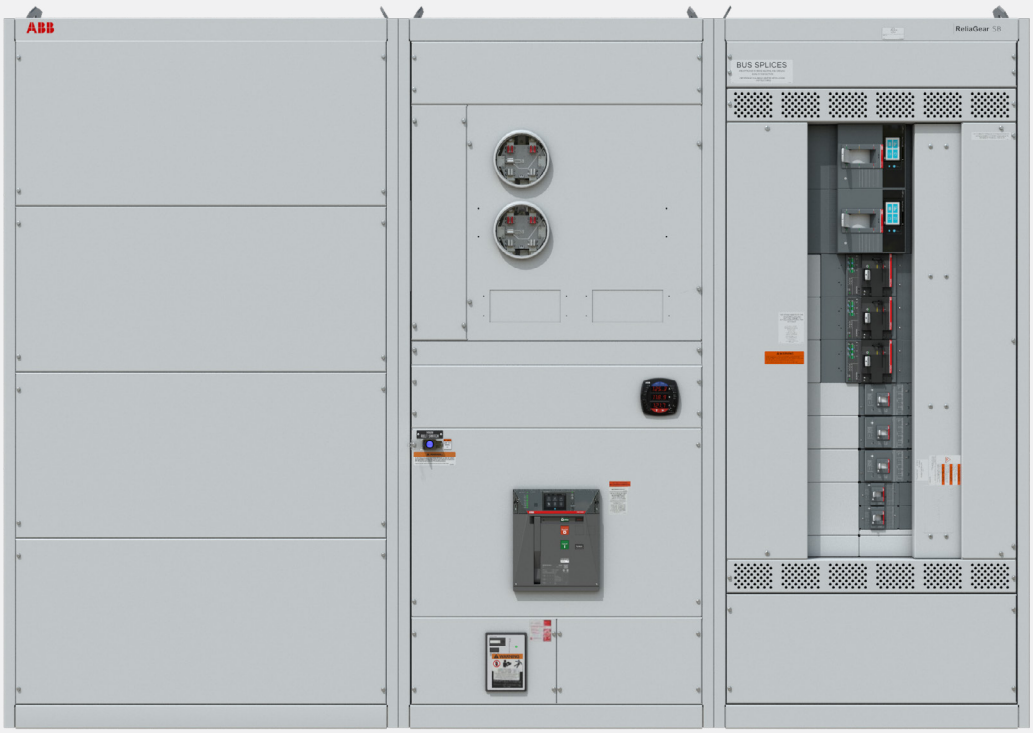


DESIGN GUIDE

RELIAGEAR® SB SWITCHBOARD – EXCELERATE

Configured to order for faster lead times

ENGINEERED
TO OUTRUN

Get the switchboard you need — faster, easier and built to fit your space. With 6000+ configuration options, ReliaGear® SB Switchboard – EXcelerate eliminates engineering delays, ensuring immediate drawing release and faster delivery. Save time and space with a compact footprint and integrated utility and main circuit breaker compartments. Installation is quick and safe, thanks to plug-in SACE® Tmax® XT circuit breakers and a finger-safe bus stack. Designed for growth and efficiency, modular sections allow for seamless expansion, while advanced trip units provide high-precision protection and power management. Whether you need NEMA Type 1 or Type 3R enclosures, EUSERC or other utility compartments, compliance is covered. Plus, integrated power metering gives you real-time energy insights, helping you optimize efficiency and cut costs.

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01
LAYOUT, SIZING AND DIMENSIONS

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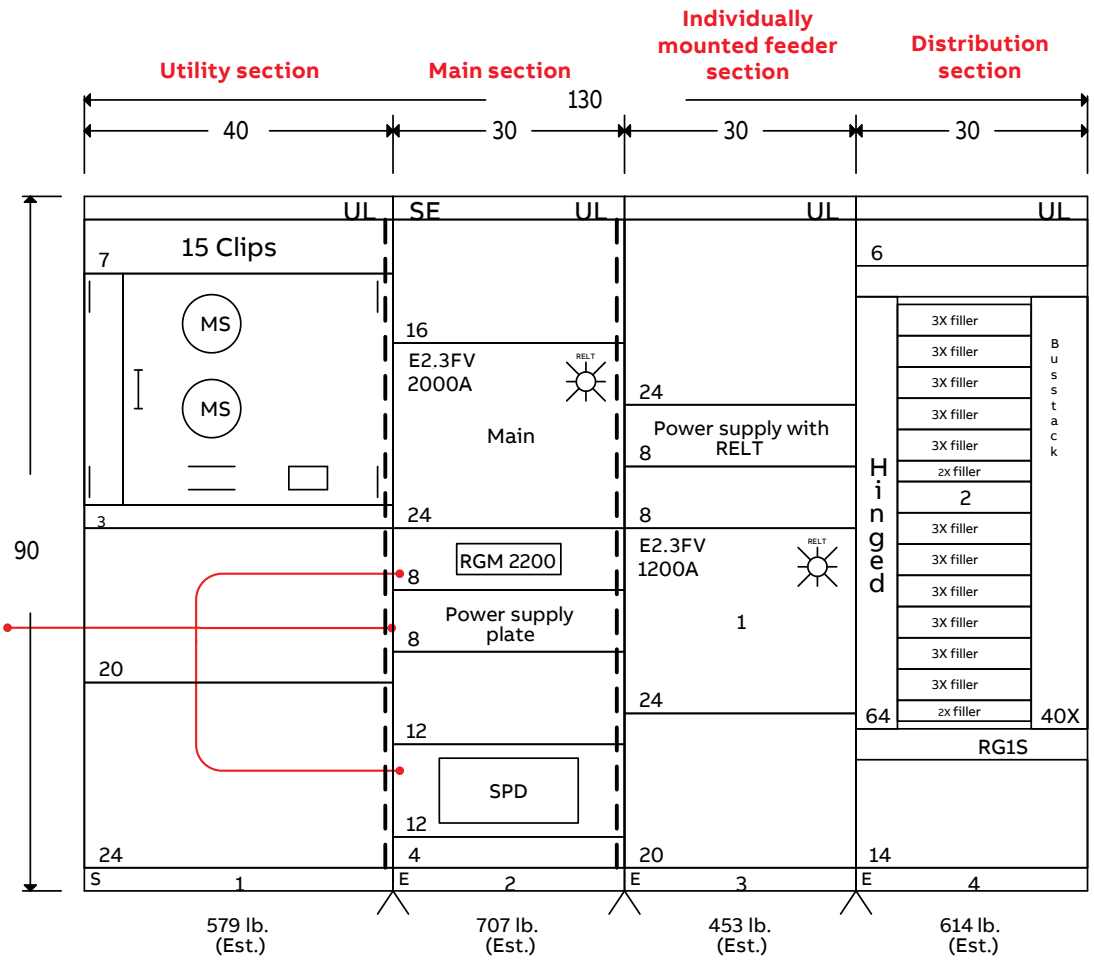
01

LAYOUT, SIZING AND DIMENSIONS

Typical layout of the EXcelerate switchboard and selection steps

01.1 Sections for application

01 Typical example of EXcelerate switchboard layout (modular design of sections)



Auxiliary circuits always in a standard location. If not selected, a blank cover will be placed in the location.

EXCELERATE SWITCHBOARD DESIGN SELECTION STEPS

Step	Selection
01.1	Select sections for application
01.2	Select NEMA Type 1 or Type 3R enclosure
01.3	Select main section
01.4	Select connection type
01.5	Select feeder distribution section
01.6	Select utility compartment
02.2	Select circuit breakers
03.1	Select additional accessories

Layout, sizing and dimensions

01.2 NEMA Type 1 or NEMA Type 3R enclosure

NEMA Type 1 to NEMA Type 3R enclosures (non-walk-in) layout, sizing and dimensions changes/differences explained

Layout B



NEMA Type 1



NEMA Type 3R
single-door
W ≤35"



NEMA Type 3R
double-door
W >35"

NEMA ENCLOSURE DIMENSIONS

Enclosure type	W	D	H
NEMA 1	Determined by amperage	35"	90"
NEMA 3R	Determined by amperage	40" + 5" roof overhang	91"
NEMA 3R with EUSERC utility	Determined by amperage	46" + 5" roof overhang	91"

Layout, sizing and dimensions

01.3 Main section

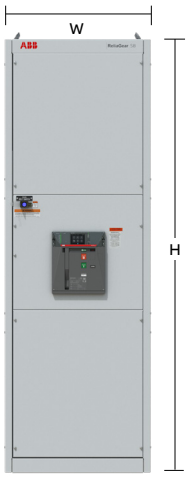
Main disconnect (cable in/cable out section), individually mounted main (IMM)
Individually mounted main disconnect section in a single section without feeder circuit breakers. Possible configuration with: a) incoming top feed lugs/cable type and outgoing lugs/cable feed type bottom load exit and b) incoming bottom feed lugs/cable type and outgoing lugs/cable feed type top load exit. It is available in NEMA Type 1 and Type 3R enclosure type with bus rating as fully rated (without bus extension).

- NEMA 1 — H = 90", see Appendix conduit figure 3
- NEMA 3R — H = 91", see Appendix conduit figure 3A

Main disconnect (fully rated with bus extension)
Individually mounted main disconnect sections designed with cross bus for connecting to varieties of feeder sections. Available with top or bottom lugs/cable incoming feed. It is available in NEMA Type 1 and Type 3R enclosure type with bus rating as fully rated (with bus extension).

- NEMA 1 — H = 90", see Appendix conduit figure 4
- NEMA 3R — H = 91", see Appendix conduit figure 4A

Layout B

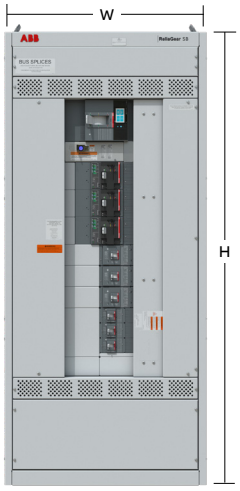


Main section — main disconnect (only section, fully rated without bus extension) or (fully rated with bus extension)

Main disconnect (group-mounted, only for 1200 A as XT7 main built-in distribution, fully rated with bus extension)
Group-mounted main disconnect designed to save space allowing main-disconnect within distribution sections and cross bus for connecting to a variety of additional feeder sections. Available with top or bottom lugs/cable incoming feed. It is available in NEMA Type 1 and Type 3R enclosure type with bus rating as fully rated (with bus extension).

- NEMA 1 — H = 90", see Appendix conduit figure 5
- NEMA 3R — H = 91", see Appendix conduit figure 5A

Layout C



Main section — main disconnect (group-mounted, only for 1200 A as XT7 main built-in distribution, fully rated with bus extension)

Layout, sizing and dimensions

01.4 Connection type

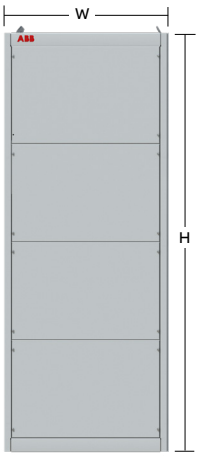
Main lug section — MLO (main lug only)
A main lug section has a lug strap and a cross bus that connects to the adjacent feeder section bus. This allows for enhanced configurability when connecting incoming cables for feeder-only lineups.

- NEMA 1 — H = 90", see Appendix conduit figure 1
- NEMA 3R — H = 91", see Appendix conduit figure 1A

Bussed pull section
A bussed pull section has a cross bus that connects to the adjacent main section bus. This allows for enhanced configurability and additional space when connecting incoming cables to the main device.

- NEMA 1 — H = 90", see Appendix conduit figure 2
- NEMA 3R — H = 91", see Appendix conduit figure 2A

Layout A



Main lug section — MLO
Bussed pull section

Blank pull section — 15" and 30" wide
Blank pull sections are available in 15- and 30-inch widths. The blank pull section provides space for pulling and installing cables. A barrier is provided for service entrance applications to meet NEC requirements. Ungrounded busing and lugs are not provided. The 15-inch blank pull section can be added to a lineup from the layout screen in empower.

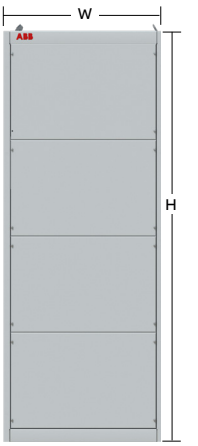
30" wide

- NEMA 1 — H = 90", see Appendix conduit figure 12
- NEMA 3R — H = 91", see Appendix conduit figure 12A

15" wide

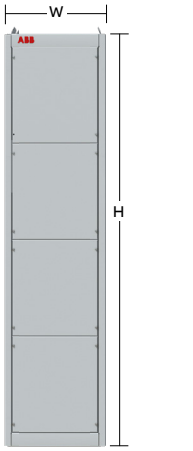
- NEMA 1 — H = 90", see Appendix conduit figure 13
- NEMA 3R — H = 91", see Appendix conduit figure 13A

Layout J



Blank pull section — 30"

Layout K



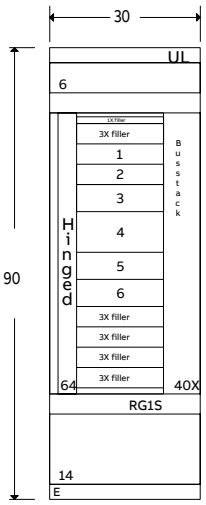
Blank pull section — 15"

Layout, sizing and dimensions

01.5 Feeder distribution section

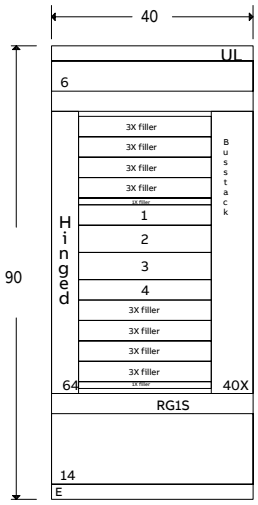
Distribution section designed to optimize footprint. The size of the section is selected automatically based on selection of circuit breakers in empower.

Small — 1 sided



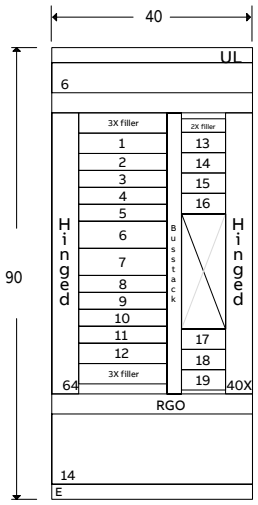
35" deep
For conduit area, see
Appendix figure 6 (NEMA 1)
or 6A (NEMA 3R)

Medium — 1 sided



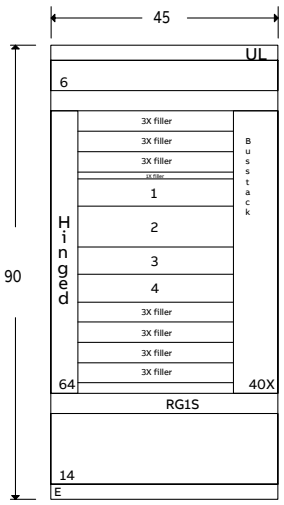
35" deep
For conduit area, see
Appendix figure 7 (NEMA 1)
or 7A (NEMA 3R)

Medium — offset



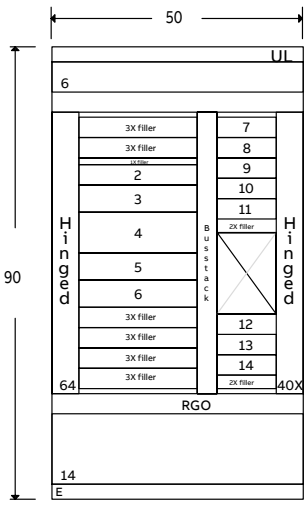
35" deep
For conduit area, see
Appendix figure 8 (NEMA 1)
or 8A (NEMA 3R)

Large — 1 sided



35" deep
For conduit area, see
Appendix figure 9 (NEMA 1)
or 9A (NEMA 3R)

Extra large — offset



35" deep
For conduit area, see
Appendix figure 10 (NEMA 1)
or 10A (NEMA 3R)

DIMENSIONS AND CRITERIA OF VARIATIONS OF GROUP-MOUNTED FEEDER DISTRIBUTION SECTION

Maximum group-mounted distribution circuit breaker frame	Lineup, incoming ampacity (A)	Maximum size/ limitation of cables per phase	Number of X spaces used	One-sided/offset	Section height (in.)	Min. section width (in.)	Section depth (in.)
XT4	≤ 2000	Any cable offered	≤ 40X	One-sided	90	30	35
XT7	1200–4000	4x 500 kcmil	≤ 40X	One-sided	90	40	35
XT7	1200–4000	3x 750 kcmil	≤ 40X	One-sided	90	45	35
XT5	1200–4000	2x 500 kcmil	40X +	Offset	90	40	35
XT7	1200–4000	3x 750 kcmil	40X +	Offset	90	50	35

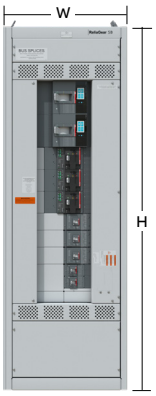
Layout, sizing and dimensions

01.5 Feeder distribution section

Distribution (group-mounted), small, one-sided (30" wide)
Distribution section with group-mounted feeder circuit breakers designed to optimize footprint in multiple feeder circuit breakers configuration. Size of the group-mounted distribution panel optimized based on selection of the critical items in empower incoming ampacity up to 2000 A. Type, ampere ratings, quantities of the distribution circuit breakers within available 40x space.

- NEMA 1 — H = 90", see Appendix conduit figure 6
- NEMA 3R — H = 91", see Appendix conduit figure 6A

Layout D

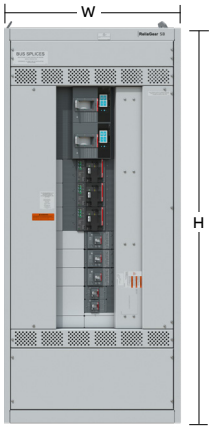


Feeder distribution section — group-mounted, small, one-sided 30"

Distribution (group-mounted), medium, one-sided (40" wide)
Distribution section with group-mounted feeder circuit breakers designed to optimize footprint in multiple feeder circuit breakers configuration. Size of the group-mounted distribution panel optimized based on selection of the critical items in empower incoming ampacity up to 4000 A. Type, ampere ratings, quantities of the distribution circuit breakers within available 40x space.

- NEMA 1 — H = 90", see Appendix conduit figure 7
- NEMA 3R — H = 91", see Appendix conduit figure 7A

Layout E

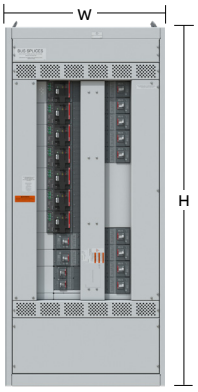


Feeder distribution section — group-mounted, medium, one-sided 40"

Distribution (group-mounted), medium, offset (40" wide)
Distribution section with group-mounted feeder circuit breakers designed to optimize footprint in multiple feeder circuit breakers configuration. Size of the group-mounted distribution panel optimized based on selection of the critical items in empower incoming ampacity up to 4000 A. Type, ampere ratings, quantities of the distribution circuit breakers available beyond 40x space.

- NEMA 1 — H = 90", see Appendix conduit figure 8
- NEMA 3R — H = 91", see Appendix conduit figure 8A

Layout F



Feeder distribution section — group-mounted, medium, offset 40"

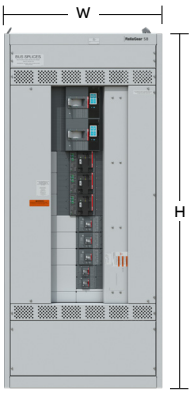
Layout, sizing and dimensions

01.5 Feeder distribution section

Distribution (group-mounted), large, one-sided (45" wide)
Distribution section with group-mounted feeder circuit breakers designed to optimize footprint in multiple feeder circuit breakers configuration. Size of the group-mounted distribution panel optimized based on selection of the critical items in empower incoming ampacity up to 4000 A. Type, ampere ratings, quantities of the distribution circuit breakers within available 40x space.

- NEMA 1 — H = 90", see Appendix conduit figure 9
- NEMA 3R — H = 91", see Appendix conduit figure 9A

Layout G

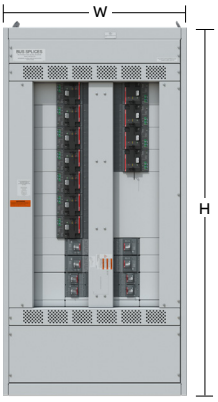


Feeder distribution section — group-mounted, large, offset 45"

Distribution (group-mounted), extra-large, offset (50" wide)
Distribution section with group-mounted feeder circuit breakers designed to optimize footprint in multiple feeder circuit breakers configuration. Size of the group-mounted distribution panel optimized based on selection of the critical items in empower incoming ampacity up to 4000 A. Type, ampere ratings, quantities of the distribution circuit breakers available beyond 40x space.

- NEMA 1 — H = 90", see Appendix conduit figure 10
- NEMA 3R — H = 91", see Appendix conduit figure 10A

Layout H

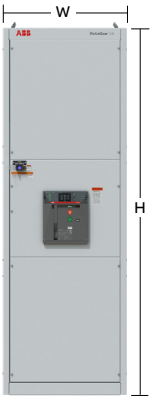


Feeder distribution section — group-mounted, extra- large, offset 50"

Individually mounted
Distribution section with individually mounted feeder circuit breakers designed for larger feeder loads. Size of the individually mounted distribution section optimized based on selection of the critical items in empower configuration for ampacity of the feeder circuit breaker.

- NEMA 1 — H = 90", see Appendix conduit figure 11
- NEMA 3R — H = 91", see Appendix conduit figure 11A

Layout I



Feeder distribution section — individually mounted

Layout, sizing and dimensions
01.6 Utility sections

EUSERC utility sections	14
Other utility sections	16

The list of EUSERC and other utility companies is continually updated. Please refer to empower or scan the QR code below to access the latest list.

Utility Compartments for ReliaGear® SB Switchboard

Scan QR code to view utility selections,
filterable by company, state and switchboard type.

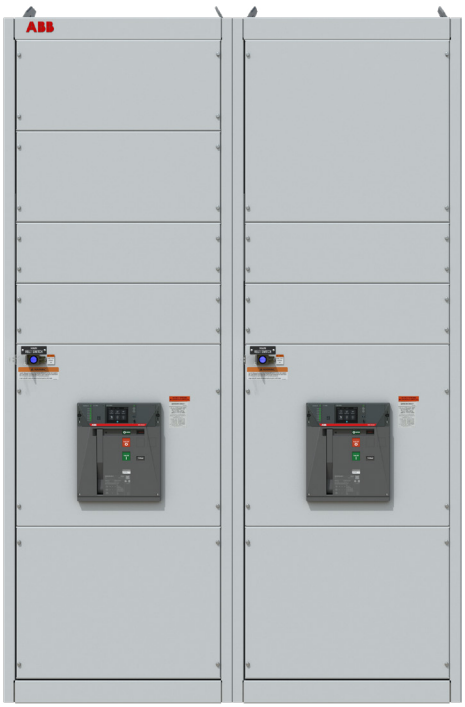
Layout, sizing and dimensions

01.6 Utility section — EUSERC (hot sequence)

Front-access EUSERC utility compartment details

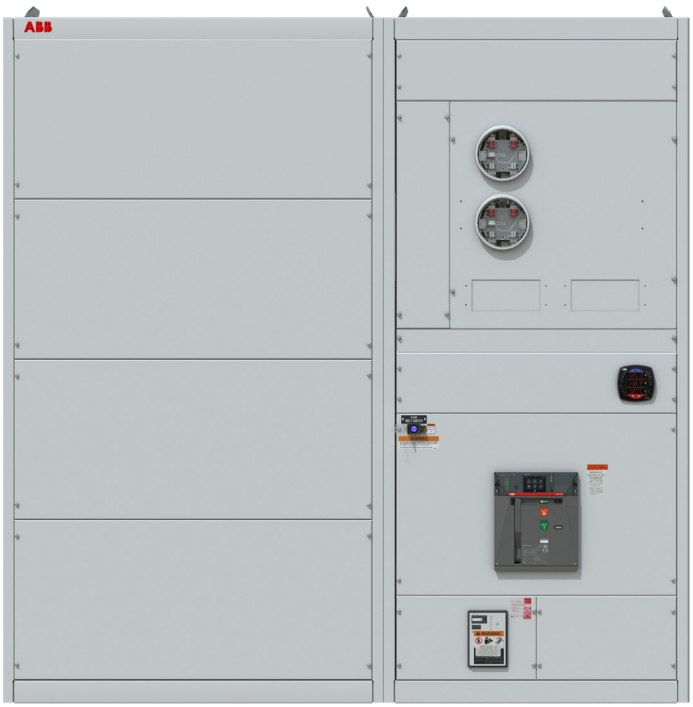
All EUSERC utility compartments are hot sequence, i.e., the main disconnect is positioned after the meter compartment. EUSERC utility compartments are available for up to 4000 A with selected utility companies. See EUSERC and other utility section table. EUSERC utility compartments or sections are available at 208/120 V and 480/277 V with bottom feed only for layout with main disconnect lineup.

Layout L



EUSERC utility section and built-in pull compartment with separate main section with main disconnect

Layout M



Separate EUSERC pull section and EUSERC utility section built-in main section with main disconnect

Layout, sizing and dimensions

01.6 Utility section — EUSERC (hot sequence)

EUSERC utility section with built-in pull compartment

All EUSERC utility compartments are hot sequence, i.e., the main disconnect is positioned after the meter compartment. EUSERC utility compartment is available for up to 2000 A. See EUSERC and other utility section table. EUSERC utility compartments or sections are available for up to 480 V maximum with bottom feed only for layout with main disconnect lineup.

A half-height bussed pull compartment combined with utility metering compartment in single section with cross bus connecting to the adjacent main section. It is available in NEMA Type 1 and Type 3R enclosures with bus rating as fully rated (with bus extension).

- NEMA 1 — H = 90", see Appendix conduit figure 14
- NEMA 3R — H = 91", see Appendix conduit figure 14A
- For 1200 to 2000 A with separate main section, refer to layout B

Layout L



EUSERC utility section and built-in pull compartment with separate main section with main disconnect, bottom feed

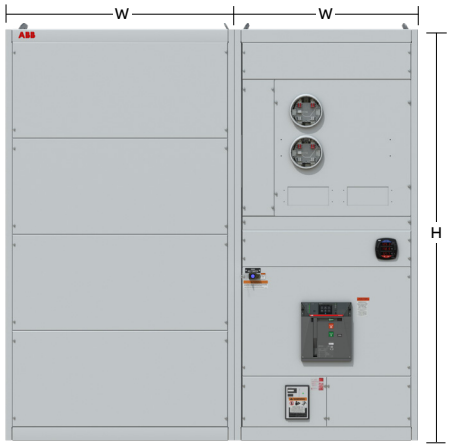
EUSERC utility section with built-in main section — main disconnect and separate EUSERC pull section

All EUSERC utility compartments are hot sequence, i.e., the main disconnect is positioned after the meter compartment. EUSERC utility compartment is available for up to 4000 A. See EUSERC and other utility section table. EUSERC utility compartments or sections are available for up to 480 V maximum with bottom feed only for layout with main disconnect lineup.

EUSERC pull sections, meeting EUSERC utility requirements, are available in 50" width and 35" depth. The standard landing uses two ½" steel studs on 1¼" vertical centers per 400 A of rating. Multiple positions are separated by 2". These sections are designed for bottom entry. A barrier is provided for service entrance to meet NEC requirements.

- NEMA 1 — H = 90", see Appendix conduit figure 15
- NEMA 3R — H = 91", see Appendix conduit figure 15A

Layout M



Separate EUSERC pull section with combined EUSERC utility section built-in main section with main disconnect, bottom feed

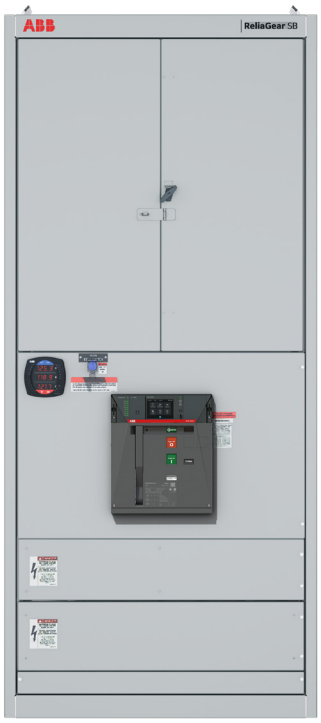
Layout, sizing and dimensions

01.6 Utility section — other (hot and cold sequence)

Front access for other utility compartment details

Other utility compartments are hot or cold sequence depending on the configuration. Other utility compartments are available for up to 4000 A for bottom feed only layout with main disconnect lineup. Other utility compartments and sections are available up to 600 V.

Layout N



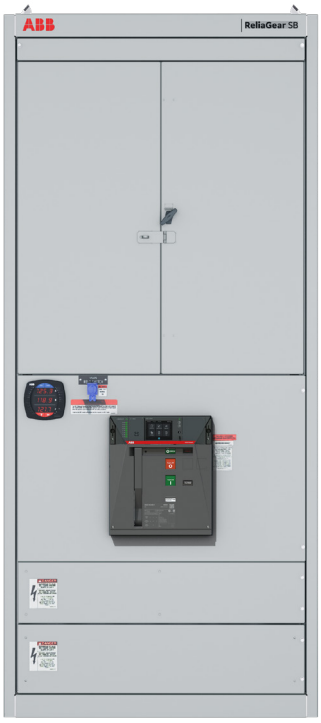
Hot sequence utility section with main disconnect with built-in pull compartment

Layout O



Hot sequence utility section and built-in pull compartment with separate main disconnect main section

Layout P



Cold sequence utility section with main disconnect with built-in pull compartment

Layout, sizing and dimensions

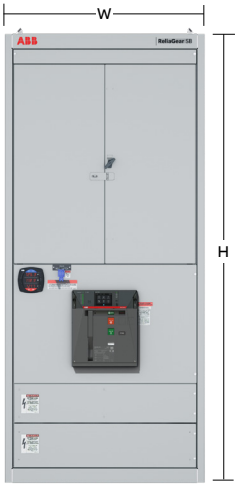
01.6 Utility section — other (hot sequence)

Other utility hot sequence section with main disconnect and built-in pull compartment
Hot sequence utility compartment with main disconnect and cable pull compartment located within the same section. Available with up to 3000 A main devices for select utility companies and bottom feed only layout.

This section is available in NEMA Type 1 and Type 3R enclosures with bus rating as fully rated (with bus extension).

- NEMA 1 — H = 90", see Appendix conduit figure 16
- NEMA 3R — H = 91", see Appendix conduit figure 16A

Layout N



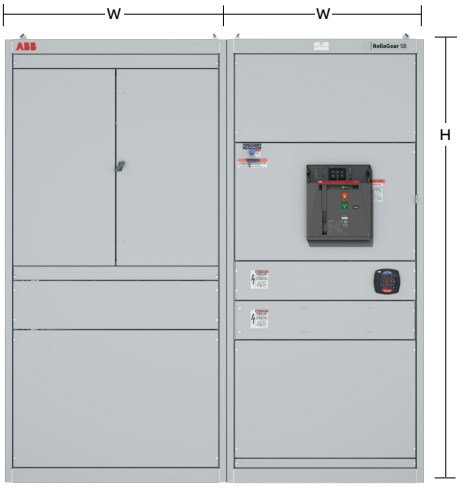
Hot sequence utility section with main disconnect with built-in pull compartment

Other utility hot sequence section with built-in pull compartment
Hot sequence utility and cable pull compartments located within the same section. The utility compartment is available up to 4000 A main device for selected utility companies and bottom feed only layout with main disconnect lineup.

This section is configured with cross bus that connects to the adjacent main section. It is available in NEMA Type 1 and Type 3R enclosures with bus rating as fully rated (with bus extension).

- NEMA 1 — H = 90", see Appendix conduit figure 17
- NEMA 3R — H = 91", see Appendix conduit figure 17A
- For 4000 A with separate main section, refer to layout B

Layout O



Hot sequence utility section and built-in pull compartment with separate main disconnect main section

Layout, sizing and dimensions

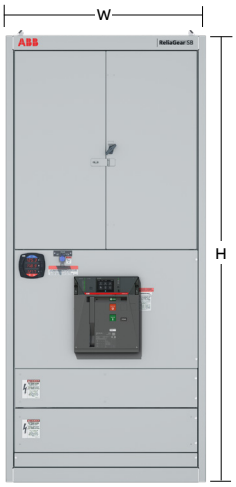
01.6 Utility section — other (cold sequence)

Other utility cold sequence section with main disconnect and built-in pull compartment
Cold sequence utility compartment with main disconnect and cable pull compartment located within the same section. Available with up to 4000 A main devices for select utility companies and bottom feed only layout.

This section is available in NEMA Type 1 and Type 3R enclosures with bus rating as fully rated (with bus extension).

- NEMA 1 — H = 90", see Appendix conduit figure 18
- NEMA 3R — H = 91", see Appendix conduit figure 18A

Layout P



Cold sequence utility section with main disconnect with built-in pull compartment



02
SWITCHBOARD PRODUCT OVERVIEW

Table of contents

Switchboard product overview

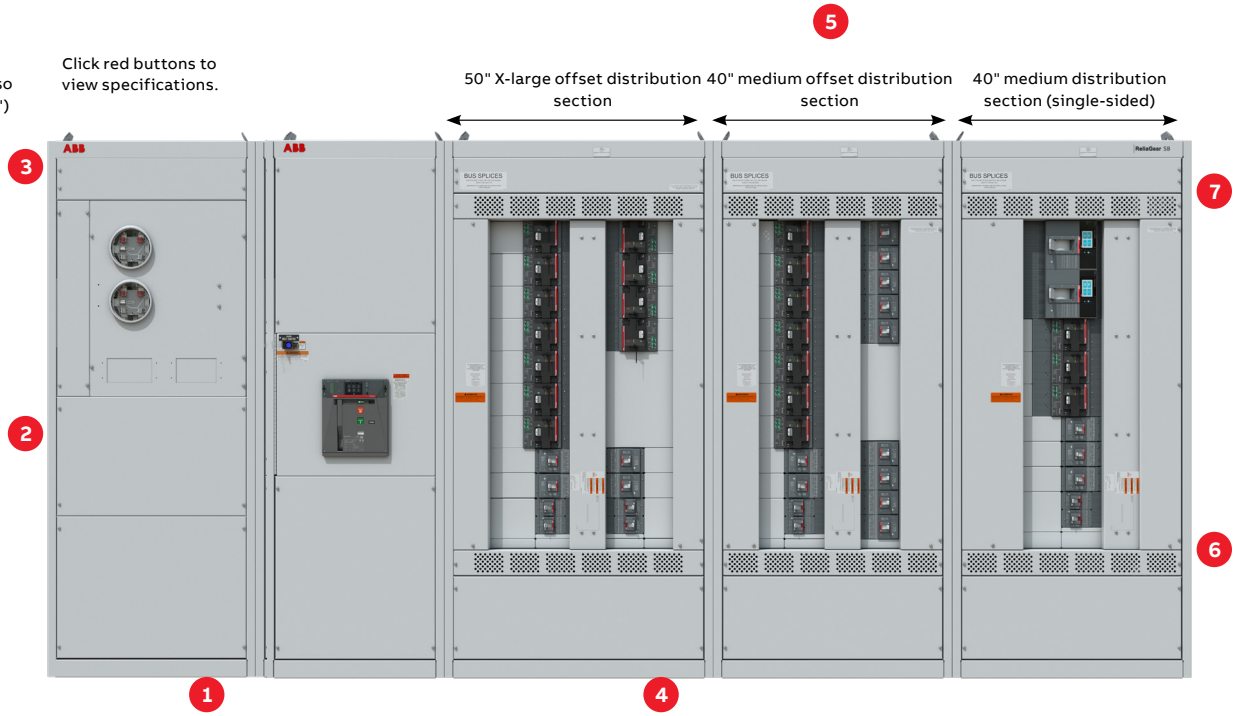
02.1 Electrical and mechanical specifications	22
02.2 ReliaGear® SB – EXcelerate designs	24
Bus and frame designs	24
Bus stack connections	24
02.3 Circuit breaker options	25
Emax 3	26
Tmax® XT	30

SWITCHBOARD PRODUCT OVERVIEW

02.1 Electrical and mechanical specifications

02 Single-sided distribution sections also available in medium (40") and large (45") width

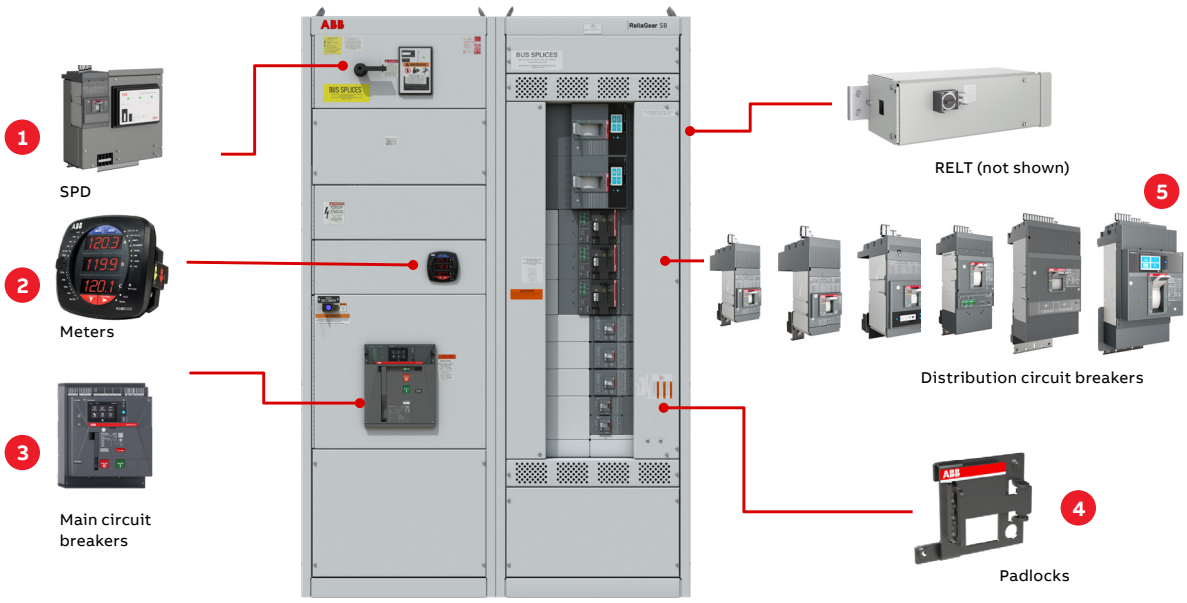
Click red buttons to view specifications.



Switchboard product overview

02.1 Electrical and mechanical specifications

Click red buttons
to view details.



03 Circuit breaker and optional accessory locations

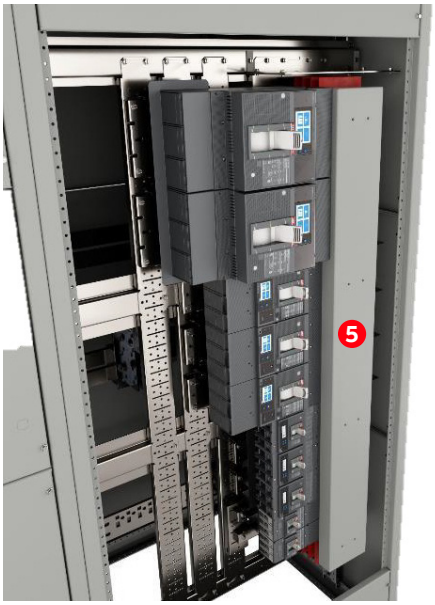
Switchboard product overview

02.2 ReliaGear® SB – EXcelerate bus and frame designs

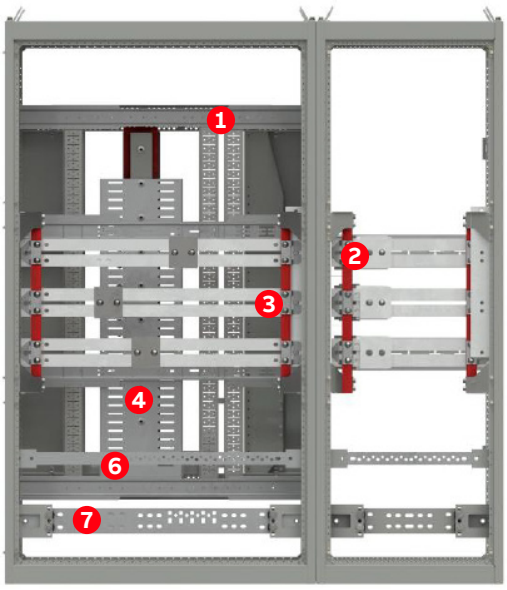
Standardized ReliaGear® SB bussing designs to drive simplification across the offering.

Standardized location of items

1. Frame
2. Main horizontal bus
3. Swing down splice plate
4. Backplane
5. Front cover (do not remove)
6. Ground
7. Neutral



Front view



Rear view

Easy to connect circuit breaker to bus stack

Plug-in connectors into bus stack for distribution sections:

- Quick to connect
  - Plug-in SACE® Tmax® XT circuit breakers (15 to 1200 A)
  - Plug-in surge protective devices
  - Plug-in RELT module
- Fast installation
- Finger-safe bus stack meets IP20 standards
- Reliable spring-loaded plug-in connectors
  - Tight connection when energized
  - Eliminating torque check
- Connect more circuit breakers and maximize circuit breaker density
- Easy circuit breaker replacement



Switchboard product overview
02.3 Circuit breaker options



04 Main circuit breaker — Emax 3



05 Feeder circuit breaker — Tmax XT

ReliaGear® SB – EXcelerate circuit breakers

SACE® Emax 3
- Fixed mount Emax 3 E2.3, E4.3, E6.3
- Short circuit ratings — S, V and H (E6.3 only)
- Frame rating — 1200, 1600, 2000, 2500, 3000, 4000 A
- Trip unit — Ekip Aware
- Reduced Energy Let Through (RELT)
- Auxiliary contacts — 4 aux (4Q) 400 V
- Padlock device, padlock in open position
- Shunt trip (optional)
- Power supply 24–240 V DC / 110–240 V AC
  - (S51/2) bell alarm — 250 V
  - (YO) opening coil — 110–120 V AC/DC (optional)

SACE® Tmax® XT
- Group-mounted Tmax XT1, XT2, XT4, XT5, XT6 and XT7
- Group-mounted in 40X panelboard interior (bus stack)
- Trip units
  - TMF (XT1)
  - TMA
  - Ekip DIP LSI
  - Ekip DIP LSIG
  - Ekip Touch (XT7)
- Padlock device, PLL fixed padlock — open (optional)

Switchboard product overview

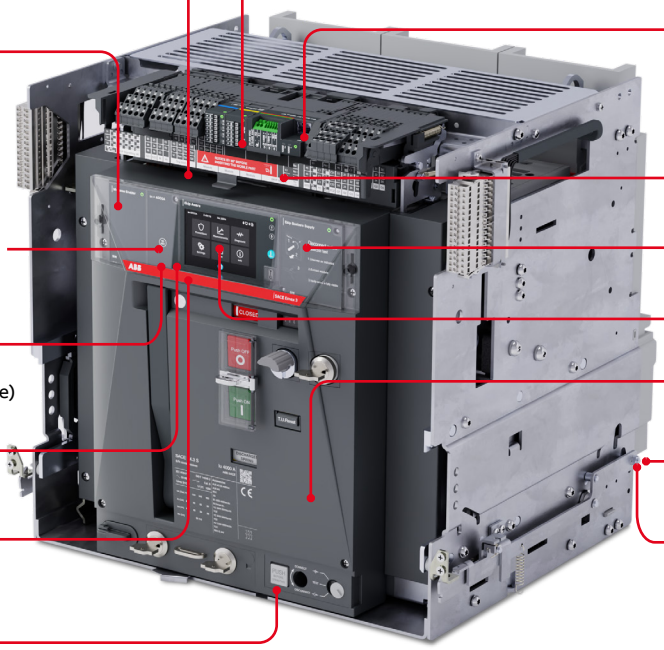
02.3 Circuit breaker options — Emax 3

SACE® Emax 3 circuit breakers

SACE Emax 3 is engineered to adapt to evolving operational demands, delivering scalable solutions that support both current performance requirements and future system expansion. Its modular design allows for seamless upgrades, ensuring long-term value and investment protection.

Designed to maximize system uptime, SACE Emax 3 incorporates advanced sensing technology alongside intelligent power and asset management capabilities. These integrated features enable continuous monitoring, predictive maintenance, and optimized system reliability—helping ensure that critical operations remain uninterrupted.

In an era where cybersecurity is essential, SACE Emax 3 is built to meet the highest security standards. It incorporates advanced protection measures aligned with IEC 62443-4-2 Security Level 2, safeguarding sensitive operational data while enhancing the safety of personnel and infrastructure.



Switchboard product overview

02.3 Circuit breaker options — Emax 3

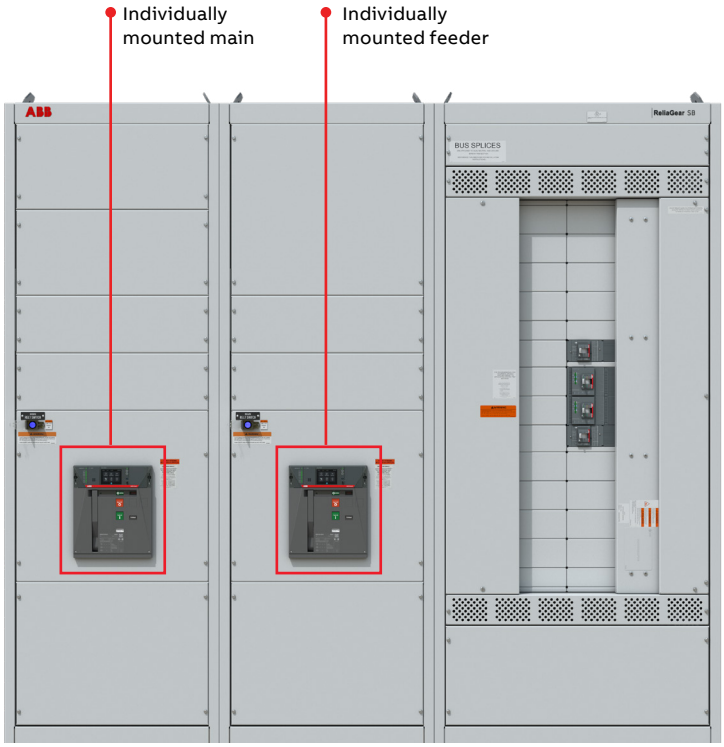


06 Emax 3 main circuit breaker

Individually mounted circuit breakers

Emax 3 circuit breaker

- UL 1066 certified, 3 poles, fixed mount only
- Ekip Aware trip unit with Reduced Energy Let Through (RELT)
- 0.5% accuracy for active power and energy
- Digital rating plug allows for configurable nominal current on customer site
- Integrated arc flash sensors, optional
- 3.5" highly readable touch panel



Switchboard product overview

02.3 Circuit breaker options — Emax 3

Emax 3 circuit breaker UL 1066

Common data		
Rated maximum voltage	[V]	635
Rated voltage	[V]	600
Test voltage (1min. 50/60 Hz)	[kV]	2.2
Frequency	[Hz]	50–60
Number of poles		3
Version[1]		Fixed (F)







SACE® Emax 3 for UL 1066		E2.3				E4.3			E6.3	
Performance levels		**–**	**S-A**	**H-A**	**V-A**	**S-A**	**H-A**	**V-A**	**H-A**	**V-A**
Current		–	–	–	–	2500	2500	–	4000	4000
		–	1200	1200	–	3200	3200	–	–	–
		–	1600	1600	–	–	–	–	–	–
		–	2000	2000	1200			2500		
		–			1600			3200		
		–			2000			–		
Interrupting rating at rated maximum voltage	254 V	–	65	85	100	65	85	100	85	100
	508 V	–	65	85	100	65	85	100	85	100
	635 V	–	65	85	85	65	85	100	85	100
Rated short time current		–	65	85	85	65	85	100	85	100
Trip times[2]	Maximum breaking time for I>Icw	–	40	40	40	40	40	40	40	40
	Maximum breaking time for I>Icw	–	25	25	25	25	25	25	25	25
Overall dimensions (inch/mm)	H - Fixed	14.61/371				14.61/371			14.61/371	
	D - Fixed	10.63/270				10.63/270			10.63/270	
	W - Fixed 3p	10.87/276				15.12/384			30.00/762	
Weights	Fixed 3p	115 lb. - 52 kg				Up to 2500 A: 161 lb. - 73 kg 3200 A: 201 lb. - 91 kg			314 lb. 142 kg	

SACE Emax 3 for UL 1066			E2.3			E4.3				E6.3
Anticipated mechanical life with regular ordinary maintenance prescribed by the manufacturer		[A]	< 1600	1600	2000	< 2500	2500	3200	3600	4000
		[No. cycles x 1000]	25	25	25	20	20	20	20	12
	Frequency	[Cycles/ hour]	60	60	60	60	60	60	60	60
Anticipated electrical life with regular ordinary maintenance prescribed by the manufacturer	508 V	[No. cycles x 1000]	15	12	10	10	8	7	7	4
	635 V	[No. cycles x 1000]	15	10	8	10	8	7	7	4
	Frequency	[Cycles/ hour]	30	30	30	20	20	20	20	10

[1] Fixed version only
[2] Total clearing time is the sum of breaking time and trip time

Switchboard product overview

02.3 Circuit breaker options – Emax 3

Emax 3 circuit breaker trip unit — Ekip Aware

Ekip Aware seamlessly integrates and evolves with your system for a scalable, future-ready solution.

One trip unit for all your needs



Upgrade with Ekip Aware

- Ekip Aware is a trip unit platform designed for extensive field customization, minimizing lead-times and installation efforts
- Customizable protection: On-demand upgrades tailored to meet specific requirements, including field-upgradable short time delay (S) and ground fault protection (G)
- Digital upgrade architecture: Adapt to fast growing design changes and ensure systems remain relevant and effective
- Digital rating plugs: Enable flexible/scalable load growth, with the ability to adjust the nominal current without the need for additional hardware

Intuitive interaction

- Refining the overall user experience, the Ekip Aware trip unit platform offers a comprehensive set of features designed for ease of use and connectivity
- Improved HMI user interface: Facilitates easier navigation and programming of key settings and parameters
- Standard USB-C connection to Ekip Connect 3 software and EPiC mobile app: Enables test and programming, in addition to providing temporary power supply for Ekip Aware trip units
- Extended wireless connectivity: Supports smartphones through NFC and Bluetooth for greater accessibility

Enhanced communication

- Recognizing the need for adaptability in an era of rapidly increasing connected devices, Emax 3 and Ekip Aware feature an optimized communication architecture that meets modern application requirements for flexibility and reliability
- Modbus TCP: Utilizes daisy chain connections to simplify cabling and streamline installation
- Redundancy: Incorporates IEC 61850, Profinet and Ethernet/IP to enhance system reliability

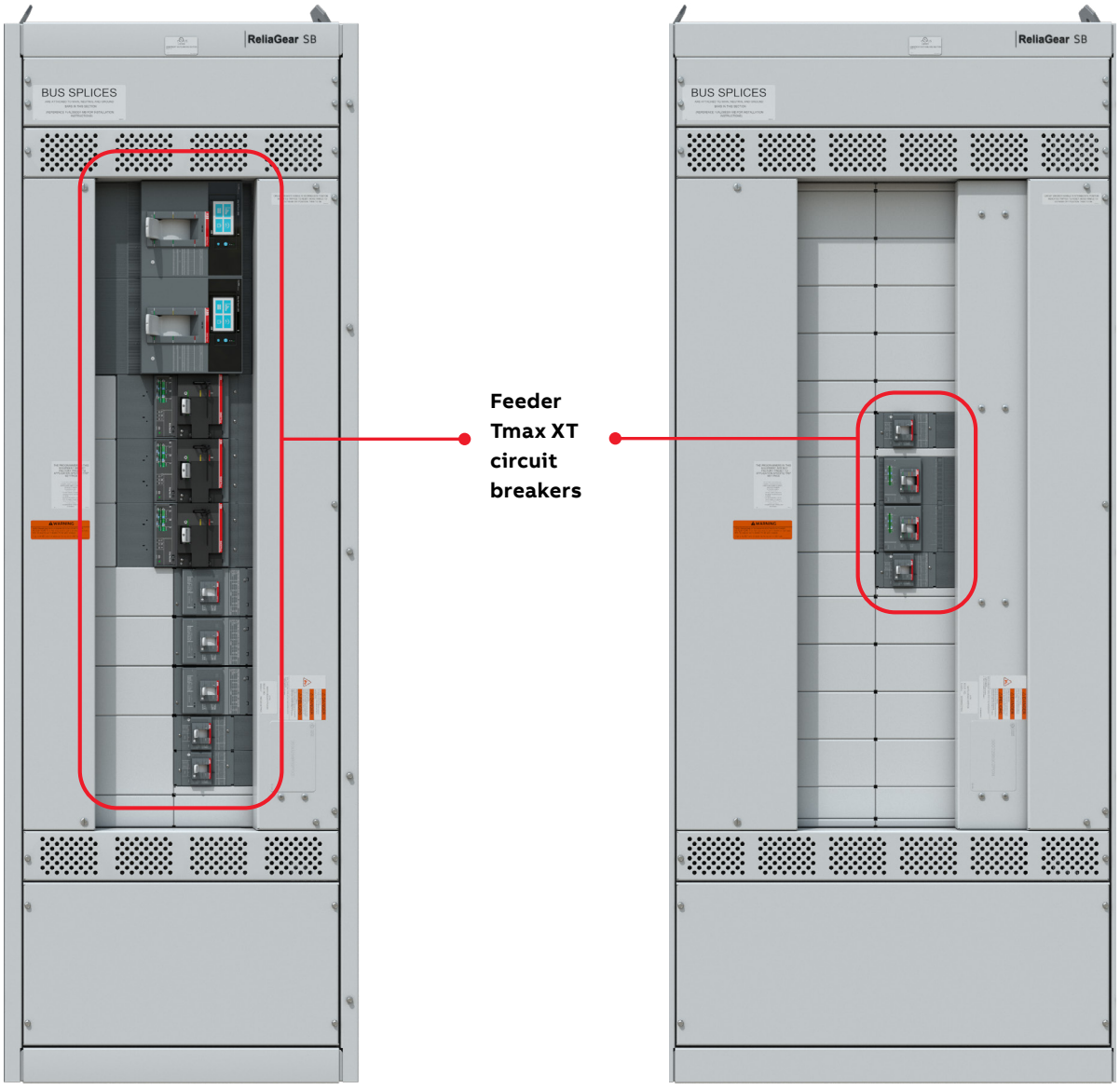
Expanded protection

- Ekip Aware offers a comprehensive offering to address various levels and scenarios to help ensure protection of personnel and assets
- Integrated arc flash: Supports up to 12 sensors per circuit breaker and the ability to reduce incident energy to less than < 8 cal/cm^2 @ 480 V up to 100 kA bolted fault, and reduces clearing time to less than 40 ms
- Advanced MDGF: This innovative solution reduces installation time and minimizes the overall number of components required, ensuring a more effective and streamlined solution
- Sensing managed via Com Link internal breaker-to-breaker communication
- No need for additional phase current transformers or external summing current transformers
- Reduced Energy Let Through (RELT): Increase operator safety during maintenance in accordance with NEC 240.87, providing clear communication and multiple methods for activation

Switchboard product overview

02.3 Circuit breaker options — Tmax® XT

Group-mounted SACE® Tmax® XT feeder circuit breakers



07 Group-mounted Tmax XT feeder circuit breakers

Switchboard product overview

02.3 Circuit breaker options — Tmax® XT

Group-mounted SACE® Tmax® XT feeder circuit breakers

Currently offered from 15 A–1200 A using Tmax® XT1 to XT7 family with various trip units.

Plug-in Tmax XT molded case circuit breakers
Each circuit breaker horizontally mounted on the bus stack is provided with a line-side connector (LSC) and a mounting bracket. The LSC is designed to ensure an easy and accurate connection between the circuit breaker and conductive busbar. A patented clip design with loaded spring ensures full contact in any circumstance. Each circuit breaker frame has a specific LSC with the right number of clips to ensure the highest performance.

Circuit breaker lugs offering
All ReliaGear® neXT circuit breakers are provided with a set of lugs on the load side. All lugs accept either copper or aluminum wires.

- Group-mounted in 40X panelboard interior (bus stack)
- Trip units:
  - TMF (XT1)
  - TMA
  - Ekip DIP LSI
  - Ekip DIP LSIG (except XT7 1200 A)
  - Ekip Touch for XT7 1200 A only
- Padlock device (optional)



08 SACE® Tmax® XT plug-in circuit breakers feature spring-loaded primary disconnects, enabling fast installation, easy replacement and reliable connection to maximize uptime.



09 Tmax XT plug-in circuit breakers can be installed in as little as 20 seconds.

Switchboard product overview

02.3 Circuit breaker options — Tmax® XT

Group-mounted SACE Tmax® XT feeder circuit breakers

The SACE® Tmax XT range offers higher performance, better protection and more precise metering than equivalent units and can handle from 15 A up to 1200 A.

Tmax XT1 (125 A max.)



Tmax XT2 (125 A max.)



Tmax XT4 (150 A max.)



Tmax XT5 (600 A max.)



Tmax XT6 (800 A max.)



Tmax XT7 (1200 A max.)



SACE® TMAX® XT MOLDED CASE CIRCUIT BREAKERS (MCCB)

			XT1			XT2								XT4						XT5				XT6			XT7	
Frame size		A	125			125								250						400–600				800			800-1000-1200	
Poles		No.	3			3								3						3				3			3	
Rated voltage		V	480 Δ (600Y/347)			600								600						600				600			600	
Versions			Fixed			Fixed								Fixed						Fixed				Fixed			Fixed	
Interrupting ratings			N	S	H	N	S	H[1]	L[1]	N	S	H[1]	L[1]	N	S	H[1]	L[1]	V	N	N	S	H[1]	L[1]	N	S	H	H	L
	240 V AC	kA	50	65	100	65	100	150	200	65	100	150	200	65	100	150	200	200	65	65	100	150	200	65	100	200	100	200
	480 V AC	kA	25	35	65	25	35	65	100	25	35	65	100	25	35	65	100	150	35	35	50	65	100	35	50	65	65	100
	600 V AC	kA	18	22	25	18	22	25	35	18	22	25	50	18	22	25	50	65	20	18	25	35	65	20	25	35	50	65
Mechanical life		No. operations	25,000			25,000								25,000						20,000				20000			10,000	
		No. hourly operations	240			240								240						240				240			240	
Dimensions – fixed (width x depth x height)		in./mm	3.0 x 7.2 x 10.4 / 77 x 184 x 265			4.1 x 7.8 x 9.8 / 105 x 198 x 249								4.2 x 7.8 x 9.8 / 105 x 198 x 249						5.6 x 8.1 x 15.2 / 141 x 206 x 387				8.3 x 8.3 x 19.2 / 210 x 211 x 488			8.3 x 10.9 x 19.2 / 211 x 277 x 488	
Weight		lb./kg	4.2 / 1.9			6.1 / 2.8								7.0 / 3.2						15.4 / 7.0				26.5 / 12.0			37.9 / 17.2	
TRIP UNITS FOR POWER DISTRIBUTION																												
TMF			•																									
TMA														•						•				•				
Ekip DIP						•								•						•				•			•	
Ekip Touch																											•	

[1] Current-limiting circuit breaker in 480 V AC and 600 V AC

Switchboard product overview

02.3 Circuit breaker options — Tmax® XT

Tmax® XT trip units

SACE® Tmax XT trip units represent a new benchmark for molded case circuit breakers, being able to satisfy any performance requirement. The Tmax XT trip units are designed to be used in a wide range of applications. These complete, flexible protection trip units can be adapted to the actual level of protection required, independently of the complexity of the system.

Three trip unit styles are available in the EXcelerate program:
- Thermal-magnetic trip unit
- Ekip DIP trip unit
- Ekip Touch trip units

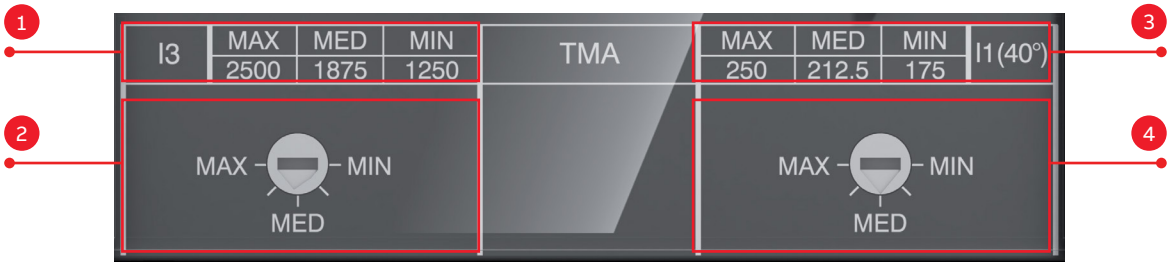
Thermal-magnetic trip units

The thermal-magnetic trip unit is an easy solution for protection against overloads and short circuits. Overload protection is enabled by the ABB thermal device, based on a temperature-dependent bimetal heated by current. Protection against short circuit is realized with a magnetic device.

There are two types of thermal-magnetic trip units:
- TMF is thermal-magnetic fixed (LI)
- TMA is thermal-magnetic adjustable

  - Adjustable thermal (L) and magnetic (I)
  - Long time adjustable between 0.7 and 1.0

Key for thermal-magnetic adjustable (TMA) trip unit:
1. Current threshold for short-circuit protection.
2. Rotary switch for short-circuit protection.
3. Current threshold for overload protection up to 105 °F (40 °C).
4. Rotary switch for overload threshold setting.



FEEDER, GROUP-MOUNTED CIRCUIT BREAKER TRIP CURRENT SETTINGS

MCCBs		In (A)																										
Circuit breaker type	Trip unit type	15	20	25	30	35	40	45	50	60	70	80	90	100	110	125	150	175	200	225	250	300	400	500	600	800	1000	1200
XT1	TMF	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•												
XT4	TMA											•	•	•	•	•	•	•	•	•	•							
XT5	TMA																					•	•	•	•			
XT6	TMA																									•		

Switchboard product overview

02.3 Circuit breaker options — Tmax® XT

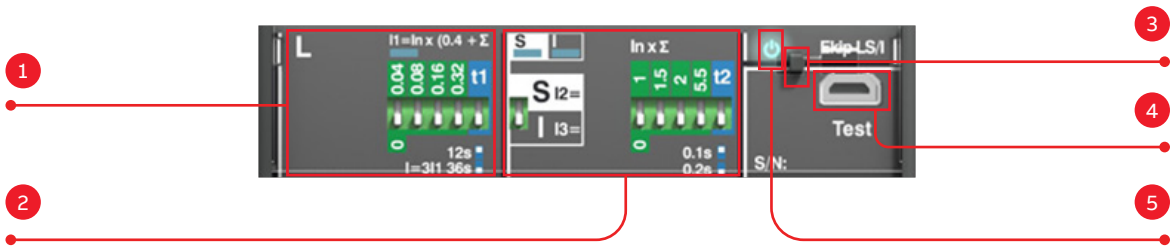
Ekip DIP trip units

The first level of electronic trip units, Ekip DIP trip units are based on microprocessor technologies and guarantee high reliability, protection adjustability and coordination. They provide protection against overloads, selective short circuits, short circuits and ground faults.

The power required for their operation is provided directly from the current sensors.

- Ekip DIP is LSI or LSIG
  - Long time adjustable between 0.4 and 1.0 in 0.02 increments

Key for Tmax XT Ekip DIP trip unit:
1. DIP switches for overload-protection setting.
2. DIP switches for short-circuit and time-delayed short circuit.
3. Slot for lead seal.
4. Test connector.
5. Power-on LED.



FEEDER, GROUP-MOUNTED CIRCUIT BREAKERS TRIP CURRENT SETTINGS

MCCBs		In (A)																											
Circuit breaker type	Trip unit type	15	20	25	30	35	40	45	50	60	70	80	90	100	110	125	150	175	200	225	250	300	400	500	600	800	1000	1200
XT2	Ekip DIP			•						•						•												
XT4	Ekip DIP													•			•				•							
XT5	Ekip DIP																						• •		•			
XT6	Ekip DIP																									•		
XT7	Ekip DIP																								•	•	•	

Circuit breaker frame	In	Minimum trip amps
XT2–125 A	25 A	10 A
	60 A	30 A
	125 A	70 A
XT4–250 A	100 A	40 A
	150 A	60 A
	250 A	100 A
XT5–400 A	300 A	120 A
	400 A	175 A
XT5–600 A	600 A	250 A
XT6–800 A	800 A	350 A
XT7–800 A	600 A	250 A
	800 A	350 A
XT7–1000 A	1000 A	400 A

Switchboard product overview

02.3 Circuit breaker options — Tmax® XT

Ekip Touch trip units

Ekip Touch trip units provide a wide series of protections and high accuracy measurements of all electrical parameters. They are intended to integrate perfectly with most common automation and supervision systems.

Key for Tmax XT Ekip Touch LSIG trip unit:
1. Power-on LED; pre-alarm LED; alarm LED.
2. Test and programming connector.
3. Touchscreen display.
4. Home pushbutton to return to homepage.
5. Pushbutton for testing and tripping information.



Trip unit	Current measurement and protection	Voltage, power, energy measurements	Voltage, power, energy protections	Embedded functions*
Ekip Touch LSI	●	○	○	○
Ekip Touch LSIG	●	○	○	○

● Default available
○ Additional features
* Please refer to the Tmax XT catalog 1SXU210248C0201 for more details.
LSIG trip units not available for single -phase applications.

FEEDER, GROUP-MOUNTED CIRCUIT BREAKER TRIP CURRENT SETTINGS

MCCBs		In (A)																										
Circuit breaker type	Trip unit type	15	20	25	30	35	40	45	50	60	70	80	90	100	110	125	150	175	200	225	250	300	400	500	600	800	1000	1200
XT7	Ekip Touch																											●

Circuit breaker frame	In	Minimum trip amps
XT7–1200 A	1200 A	500 A

More details on Tmax XT circuit breakers and trip units are available in the SACE® Tmax XT UL catalog 1SXU210248C0201 and in technical characteristic document 1SDC210199D0203.

For more details about the series ratings for these disconnect circuit breaker devices, refer to document DET008 and DET760.

03
ADDITIONAL ACCESSORIES METERING, SURGE PROTECTIVE DEVICES AND HEATERS

Table of contents

03.1 Additional accessories

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Surge protective devices	41
Space heaters	42
Line-side connectors and lugs	43

ADDITIONAL ACCESSORIES

03.1 Power metering

ReliaGear® SB Switchboard – EXcelerate is available with electronic meters RGM2000, RGM6000 and RGM7000 for high accuracy power metering.

RGM2200 digital power meter

The RGM2200 meter measures more than 40 electrical power parameters, providing a low-cost, multifunction monitoring solution for industrial and power generation applications. The RGM2200 can be easily used in a panel for generator monitoring, substation automation and more. This meter supports an optional RS485 Modbus or BACnet interface with baud rates ranging from 9600 to 57.6 k.



10 RGM2200 digital power meter

Features and benefits

- Economical meter for circuit monitoring of panels, main feeds, branch circuits and gensets
- True RMS measurement
- Universal operation — 50/60 Hz, user-programmable for medium or low voltage circuits
- Mounts in 92 mm DIN and ANSI C39.1 round cut-outs

Applications

- Monitoring of electrical loads, such as generator panels, feeders, switchgear, etc.
- Low and medium voltage applications

Monitoring and metering

- Measurements of 3-phase real-time amps, volts, power, energy, power factor and frequency
- 0.5 percent accuracy
- Optional RS485 Modbus, BACnet and KYZ pulse outputs

User interface

- Brilliant 3-line LED display
- Easy-to-use faceplate programming

Additional accessories

03.1 Power metering

RGM6000/6010 power meter

RGM6000/6010 power meter is the industry's highest performance revenue-grade meter. Based on the latest technology and an all new platform, RGM6000/6010 has a superior cost-to-performance ratio and significantly outperforms other metering products many times its price.

RGM6000/6010 can be easily used for generator monitoring, substation automation and more.

Through a built-in infrared (IrDA) port, the meter can be programmed through a laptop computer without a physical wire connection to the meter, making programming or data download convenient and safe. Using standard high speed Modbus or BACnet (RGM6010) communications, the meter can also provide data to RTUs, PLCs and other control devices. When used with ABB Ability™, the meter can be remotely monitored.

The RGM6000/6010 excels in metering energy accurately, exceeding ANSI C12.20 (0.2 percent)and IEC 687 (0.2 percent) energy measurement standards. The unit uses high speed DSP technology with high resolution A/D conversion to provide revenue certifiable accuracy for utility billing, substation metering, sub-metering and critical metering applications.

Features and benefits

- High accuracy multifunction power meter
- Superior performance at competitive pricing
- Ultra-compact, easy to install, program and use
- 0.2 percent class revenue certifiable energy and demand metering
- Fits both ANSI and DIN cut-out
- Large 3-line 0.56 inch bright LED display for better visibility and longer life
- User programmable for different system voltages and current measurements



11 RGM6000/6010 power meter

Applications

- Continuous metering of electrical loads, such as generator panels, feeders, switchgear, etc.
- Provides remote status when used with EnerVista suite of software
- Low and medium voltage applications
- Replaces multiple analog meters, saving space and installation costs

Monitoring and metering

- True RMS multifunction measurements, including voltage, current, power, frequency, energy, etc.
- Meets ANSI C12.20 (0.2 percent) and IEC 687 (0.2 percent) accuracy classes
- Future field upgradable for added functionality without removing installed meter
- Load percentage graphical bar for instant load visualization

User interface

- RS485 Modbus protocol, BACnet up to 57.6 k baud
- 3-line 0.56 inch bright red LED display
- Front IrDA port laptop communication

Additional accessories

03.1 Power metering

RGM7000 power meter

The RGM7000 power and energy meter is a low cost, powerful metering solution for large-scale deployment within an electrical distribution system. Add high accuracy, data logging, power quality functionality, extensive communication and I/O to existing or new power meter applications.



12 RGM7000 power meter

Features and benefits

High accuracy revenue metering

- ANSI C12.20 0.2 CL and IEC 61850 0.2s class accuracy certification for energy measurements
- 0.007 Hz measurement for frequency control applications
- Verify accuracy using standard energy test pulse
- Compensate for transformer line loss and CT/PT errors
- Meets New York State PSC approval for submetering

Power quality monitoring

- Record waveforms at up to 512 samples per cycle
- Up to 170 events stored before rollover
- View real-time voltage and current waveforms
- Monitor THD and record harmonic magnitudes to the 40th order
- Program up to eight (8) limits for alarm and control applications

Extensive communication options

- Standard IrDA port for configuration and direct data download
- Standard RS485 port
- Modbus ASCII/RTU/TCP, DNP3 and IEC 61850 protocols
- Up to two Ethernet and IEC 61850 ports with independent IP addresses/email on alarm/data push/embedded web servers/IP whitelisting

Data logging

- Three (3) independently programmable historical trending logs
- Record 64+ parameters per log
- I/O log, system events anti-tampering log, limit alarm log

Expandable I/O

- Optional communication and I/O can be installed in the factory or added in the field
- The meter can accept up to two (2) communication or I/O cards
- Expandable communication option cards are:
  - 100BaseT Ethernet (INP100S)
  - IEC 61850 protocol Ethernet (INP300S)
- Expandable I/O option cards are:
  - Four (4) pulse outputs, four (4) status inputs (PO1S)
  - Two (2) relay outputs/two (2) status inputs (RO1S)
  - Four (4) channel, bi-directional 0–1 mA outputs (1mAOS)
  - Four (4) channel 4–20 mA outputs (20mAOS)

Additional accessories

03.1 Surge protective devices (SPDs)

Standards

- UL 1449 5th Ed. – Listed, Open Type — Mil-Std-220C
- UL 1283 Listed EMI/RFI Filter — ANSI/NFPA 70 (Art. 242)
- UL 96A – Meets SPD criteria — ANSI/IEEE C62.41 and C62.45
- cUL Listed to UL 1449 and CSA C22.2 — ANSI/IEEE C62.62 and C62.72 IEC 61643-11

ReliaGear® RGPP Series SPDs are engineered with smart thermal disconnecting technology that safely removes overstressed components before harmful external effects arise. This preventive approach enhances safety and avoids nuisance protection device tripping or thermal events.

The core suppression system includes proprietary metal oxide varistors (MOVs) for surge energy diversion and metal film capacitors for filtering low-level transient voltages and electrical noise. The design is UL Listed as an open-type SPD, suitable for both factory and field installation or replacement.

To support ongoing monitoring, the product includes an LED status system:

- Red LED indicates complete depletion and triggers replacement
- Orange LED signals degraded but operational protection — giving the user a window to replace the SPD before total failure



13 SPD

Additionally, a rotatable LCD provides real-time feedback on key system parameters, including:

- Surge event counts
- TOV incidents
- Power outages
- Neutral-to-ground voltage over 20 V
- Remaining MOV life

This combination of protection, diagnostics and replaceability makes the RGPP Series highly suitable for demanding power distribution applications.

The RGPP Series can be installed on either side of the panel bus stack, in any enclosure width. For connection to the bus stack, a separate plug-in disconnect (ReliaGear Tmax® XT MCB) is required (see page 32). The total solution consumes 9X of bus stack space (6X for the RGPP Series SPD, 3X for the MCB disconnect).

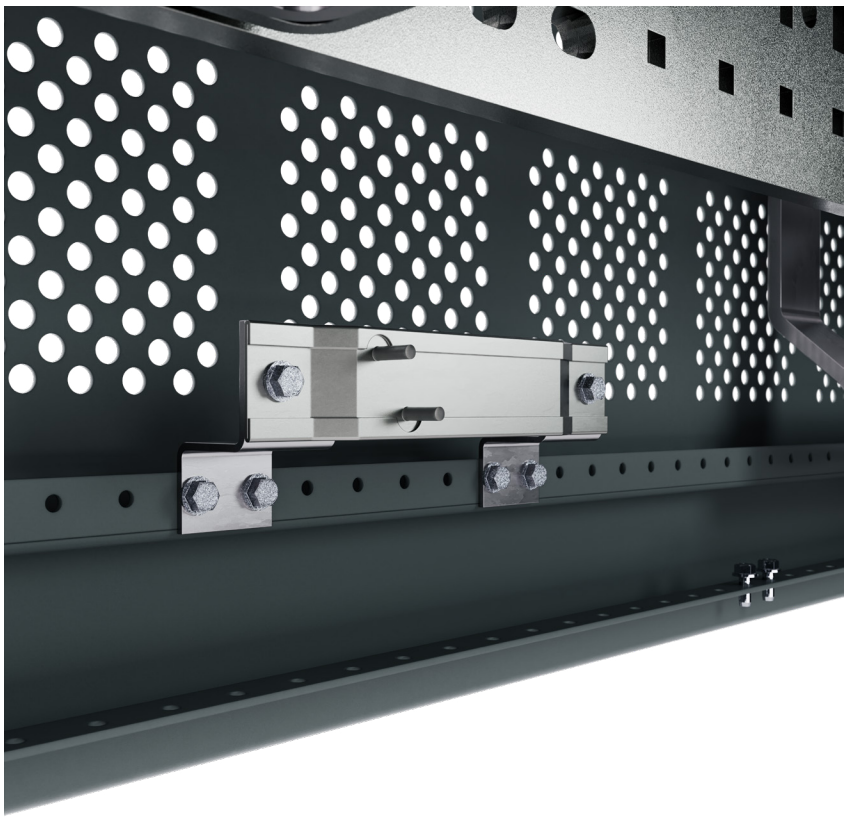
Additional accessories

03.1 Space heaters

Two different circuits are available for heaters, depending on the number of sections in the switchboard.

The first circuit is used for up to three (3) sections of the EXcelerate switchboard. The second circuit is used for switchboards with more than three (3) sections, up to a maximum of six (6) sections.

A thermostat and humidistat are wired parallel in the circuit to maintain set temperature inside the respective sections using a heater strip or series of heater strips within each section of the switchboard.



14 Heater

Additional accessories

03.1 Line-side connectors and lugs for molded case circuit breakers

Line-side connectors

Each circuit breaker horizontally mounted on the bus stack is provided with a line-side connector (LSC) and a mounting bracket. The LSC is designed to ensure an easy and accurate connection between the circuit breaker and conductive busbar. A patented clip design with loaded spring ensures full contact in any circumstance. Each circuit breaker frame has a specific LSC with the right number of clips to ensure the highest performance.

Circuit breaker lugs offering

All ReliaGear® neXT circuit breakers are provided with a set of load-side lugs. All lugs accept either copper or aluminum wires.

AVAILABLE LUGS AND CABLE RANGES FOR THE DISTRIBUTION CIRCUIT BREAKERS BASED ON CIRCUIT BREAKER FRAME AND AMPACITY

Frame	Ampacity (A)	Wire size (AWG or kcmil) Cu or Al	No. of lugs per phase/neutral	No. of cables per lug
XT1	15–25	14–1/0	1	1
XT1	30–125	10–2/0	1	1
XT2	All	10–2/0	1	1
XT4	80–125	4–300	1	1
XT4	150–250	3/0–350	1	1
XT5	All	2/0–500	1	2
XT5	All	500–750	1	2
XT6	All	2/0–400	1	3
XT6	All	500–750	1	2
XT7	All	4/0–500	1	4
XT7	All	500–750	1	3

Connectors and lugs for MLO, bussed pull sections, main disconnect sections and individually mounted feeder sections

AVAILABLE LUGS AND CABLE RANGES FOR THE MAIN LUG ONLY (MLO) OPTION, BUSSED PULL SECTIONS, MAIN DISCONNECT SECTIONS AND INDIVIDUALLY MOUNTED FEEDER SECTIONS

Ampacity (A)	Wire size (AWG or kcmil) Cu or Al	No. of lugs per phase/neutral	No. of cables per lug
1200	3/0–750	3	2
1600	3/0–750	4	2
2000	3/0–750	4	2
2500	3/0–750	5	2
3000	3/0–750	5	2
4000	3/0–750	7	2

Lugs for EUSERC utility sections

AVAILABLE LUGS AND CABLE RANGES FOR EUSERC UTILITY SECTIONS (WHEN PROVIDED)

Ampacity (A)	Wire size (AWG or kcmil) Cu or Al	No. of lugs per phase/neutral	No. of cables per lug
1200	3/0–800	3	1
1600	3/0–800	4	1
2000	3/0–800	5	1
2500	3/0–800	8	1
3000	3/0–800	8	1
4000	3/0–800	10	1

04
APPENDIX

Table of contents

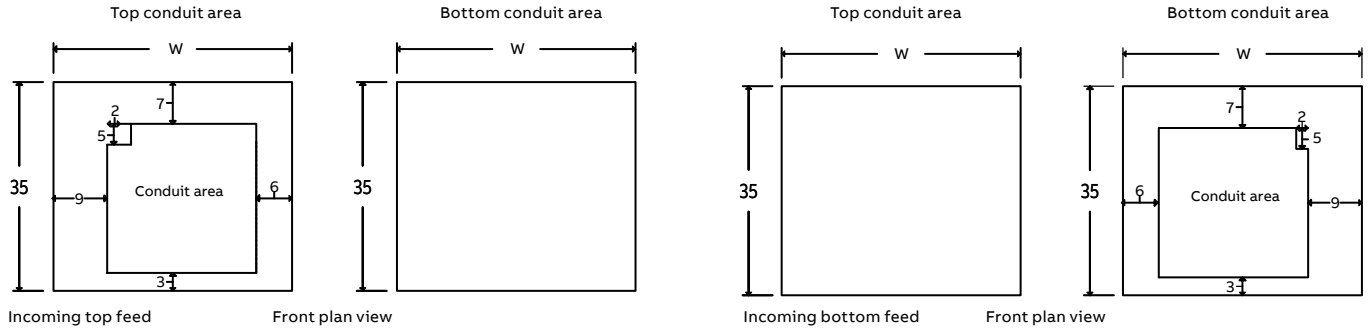
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Conduit area — NEMA Type 1 enclosures	46
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Conduit area

NEMA Type 1 enclosures

Main lug section — MLO

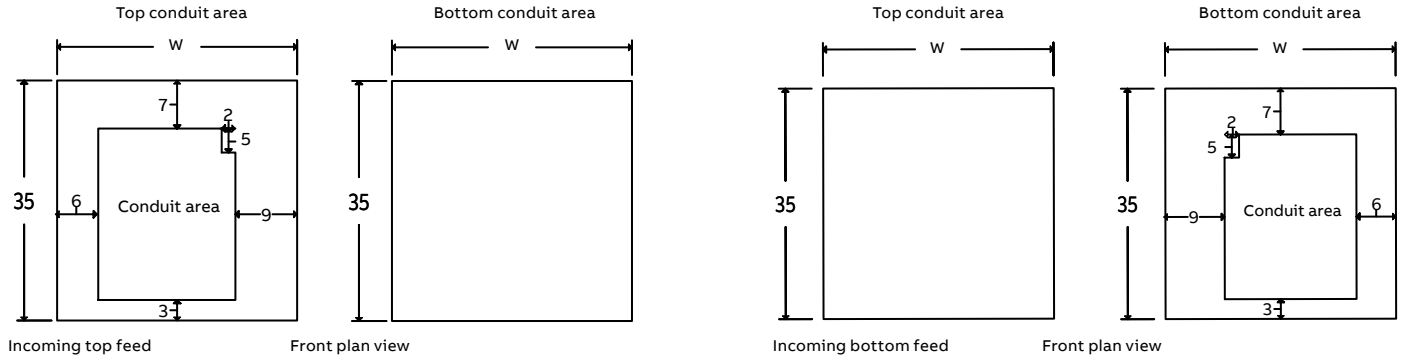
Figure 1



1200 A–3000 A: W = 35"
4000 A: W = 40"
NEMA Type 1 height: 90"

Bussed pull section

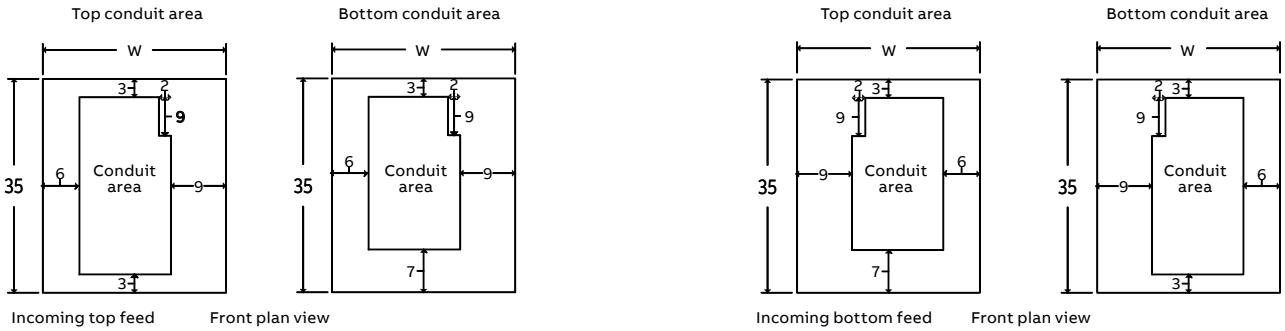
Figure 2



1200 A–3000 A: W = 35"
4000 A: W = 40"
NEMA Type 1 Height: 90"

Main section — main disconnect (ONLY section, fully rated without bus extension)

Figure 3

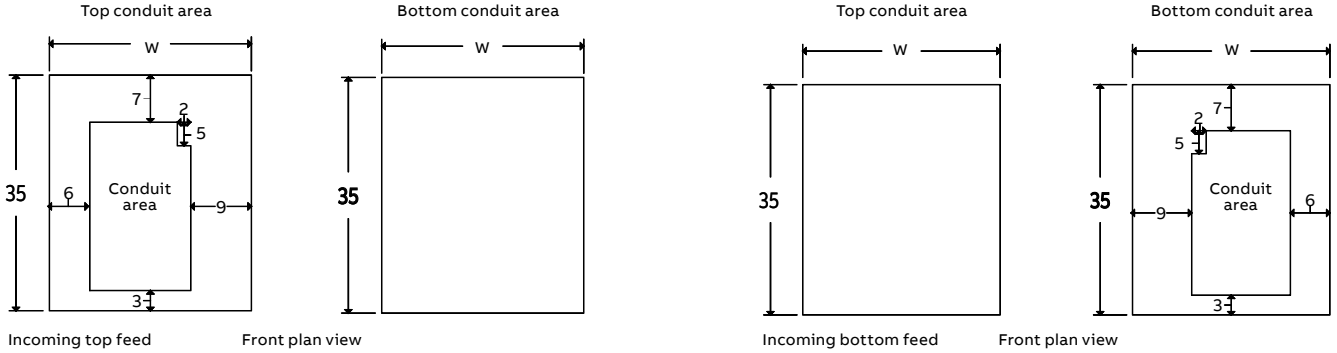


1200 A–3000 A: W = 30"
4000 A: W = 40"
NEMA Type 1 height: 90"

—

Main section — main disconnect (fully rated with bus extension)

Figure 4

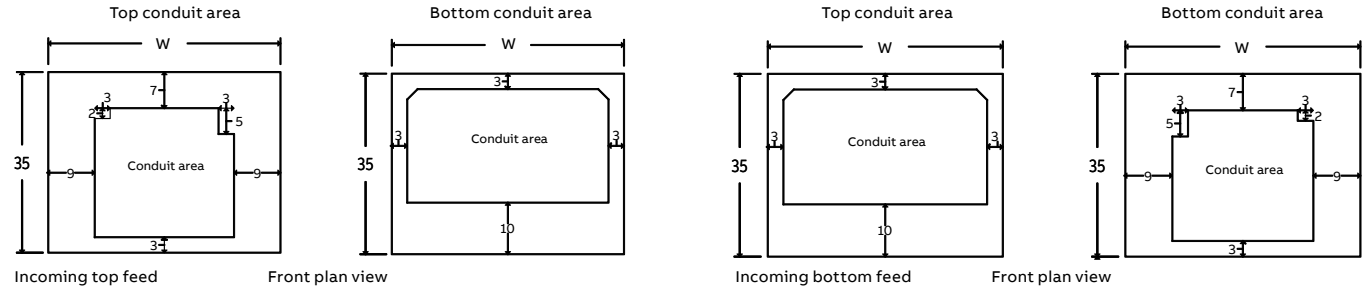


1200 A–3000 A: W = 30"
4000 A: W = 40"
NEMA Type 1 height: 90"

—

Group-mounted main section (1200 A XT7) with feeder distribution

Figure 5

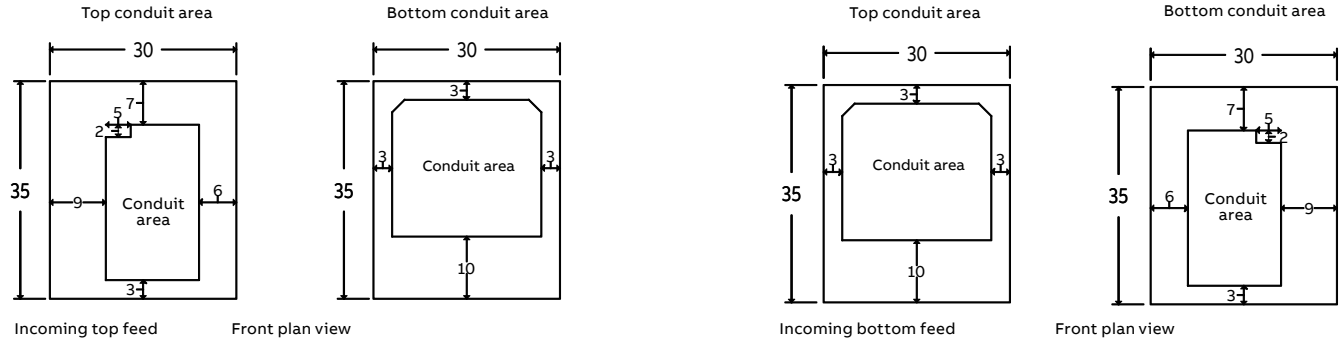


One-sided medium: W = 40"
One-sided large: W = 45"
Offset extra-large: W = 50"
NEMA Type 1 height: 90"

—

Feeder distribution section (group-mounted), small, one-sided 30" (1200 A–2000 A)

Figure 6



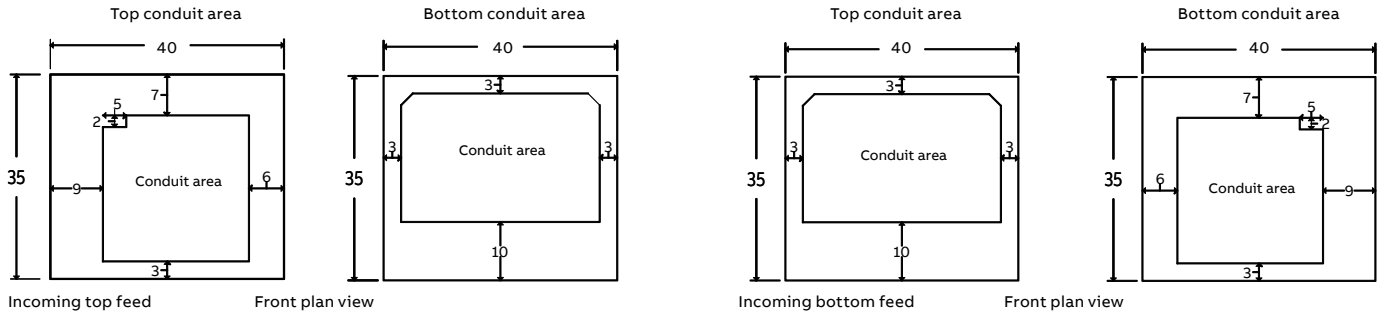
NEMA Type 1 height: 90"

Conduit area

NEMA Type 1 enclosures

Feeder distribution section (group-mounted), medium, one-sided 40"

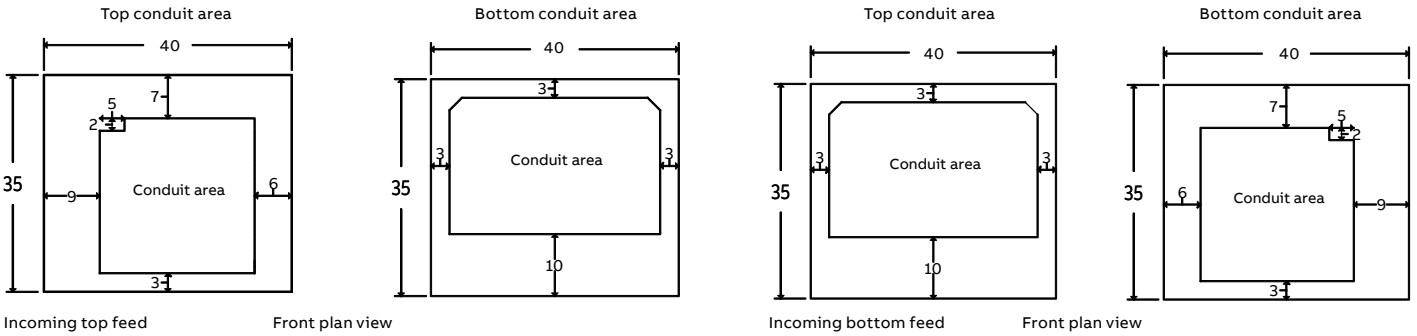
Figure 7



NEMA Type 1 height: 90"

Feeder distribution section (group-mounted), medium, offset 40"

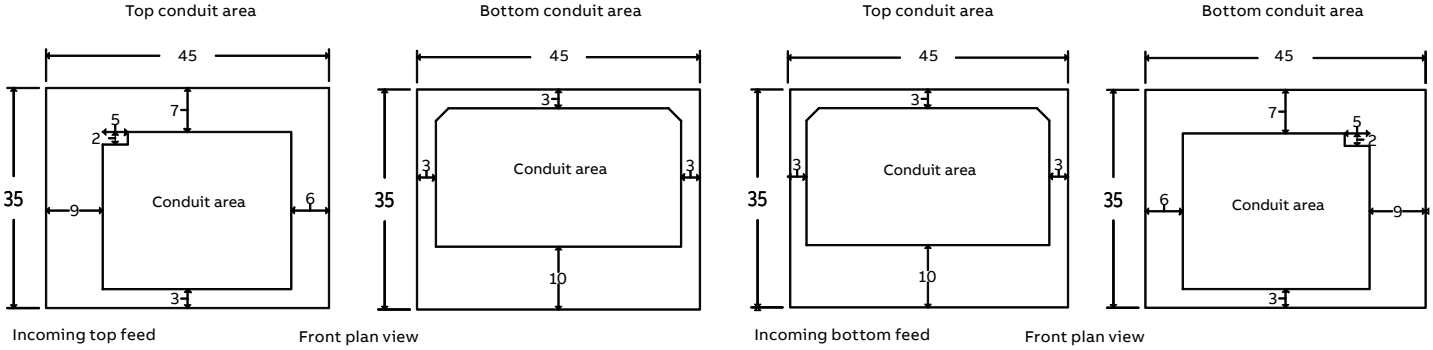
Figure 8



NEMA Type 1 height: 90"

Feeder distribution section (group-mounted), large, one-sided 45"

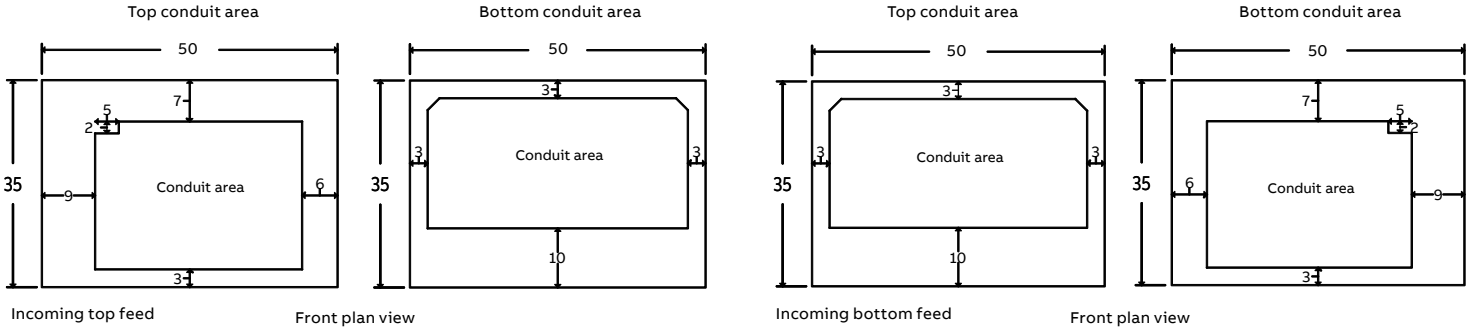
Figure 9



NEMA Type 1 height: 90"

Feeder distribution section (group-mounted), extra-large, offset 50"

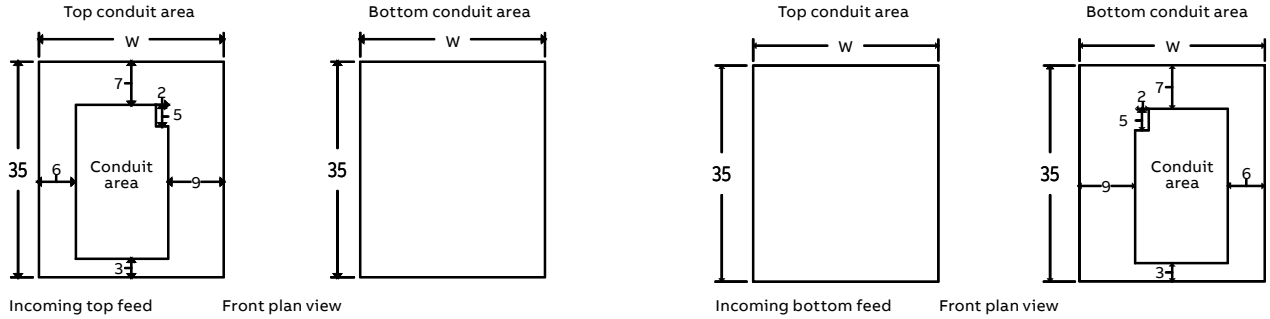
Figure 10



NEMA Type 1 height: 90"

Feeder distribution section (individually mounted)

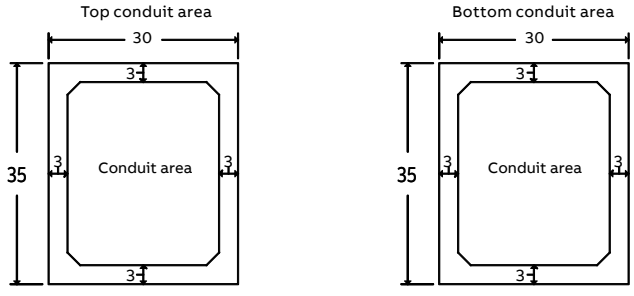
Figure 11



1200 A–3000 A: W = 30"
4000 A: W = 40"
NEMA Type 1 height: 90"

Blank pull section 30"

Figure 12



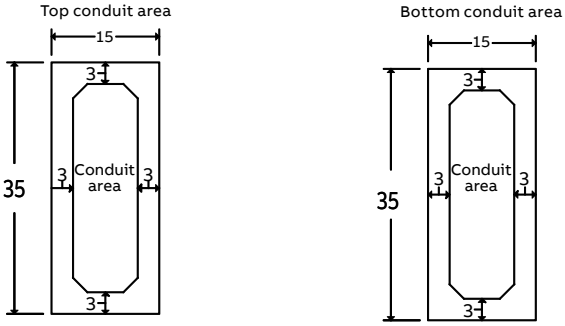
NEMA Type 1 height: 90"

Conduit area

NEMA Type 1 enclosures

Blank pull section, 15"

Figure 13



Incoming top feed
Incoming bottom feed

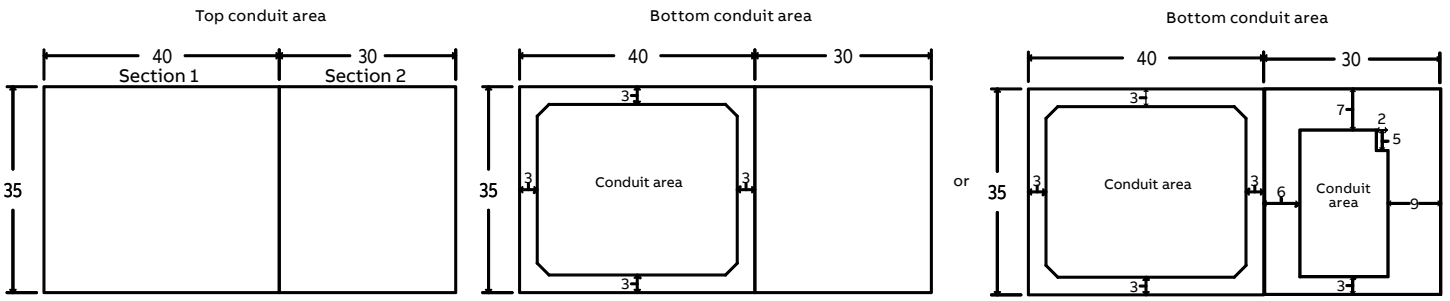
Front plan view

Outgoing top exit
Outgoing bottom exit

NEMA Type 1 height: 90"

EUSERC utility section with built-in pull compartment 1200 A–2000 A

Figure 14



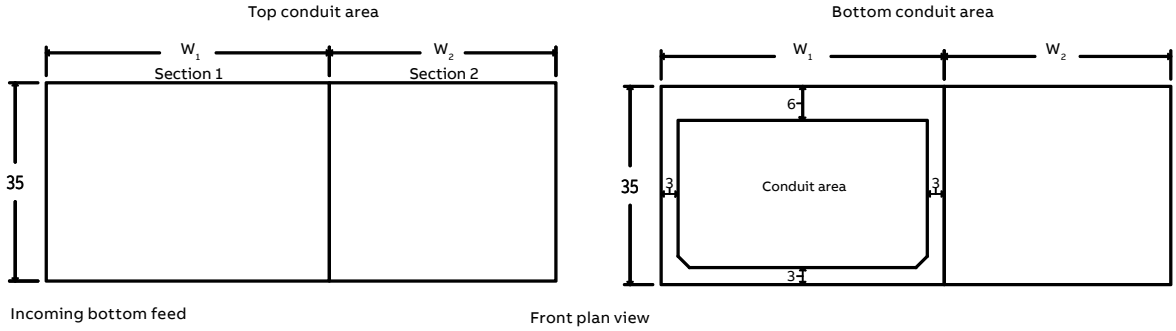
Incoming bottom feed

Front plan view

NEMA Type 1 height: 90"

EUSERC utility section with built-in main and separate EUSERC pull section

Figure 15



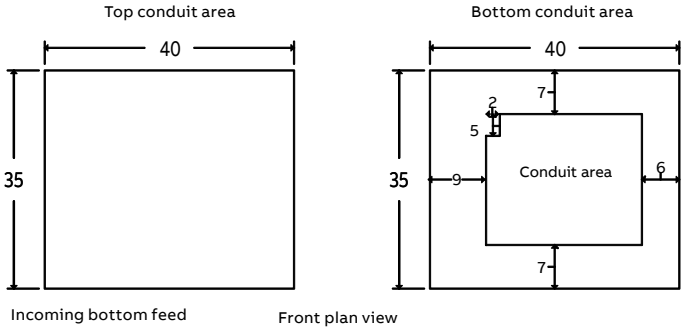
Incoming bottom feed

Front plan view

NEMA Type 1 height: 90"
1200 A–2000 A: W_1 = 40" and W_2 = 40"
2500 A–3000 A: W_1 = 50" and W_2 = 40"
4000 A: W_1 = 50" and W_2 = 50"

Other utility hot sequence utility section 1200 a–3000 a

Figure 16



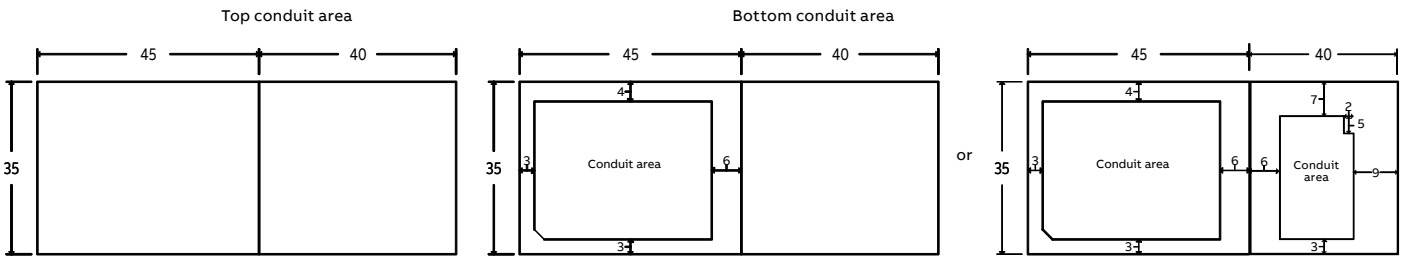
Incoming bottom feed

Front plan view

NEMA Type 1 height: 90"

Other utility hot sequence utility section 4000 a

Figure 17



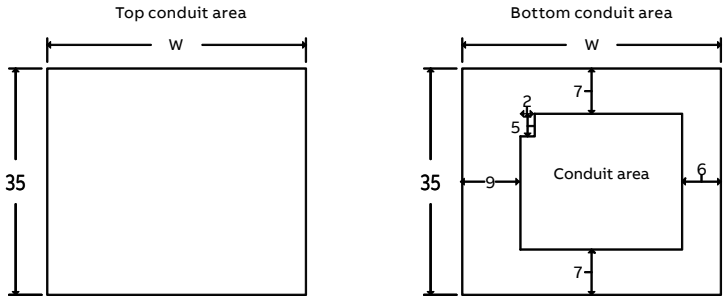
Incoming bottom feed

Front plan view

NEMA Type 1 height: 90"

Other utility (non-euserc) cold sequence utility section

Figure 18



Incoming bottom feed

Front plan view

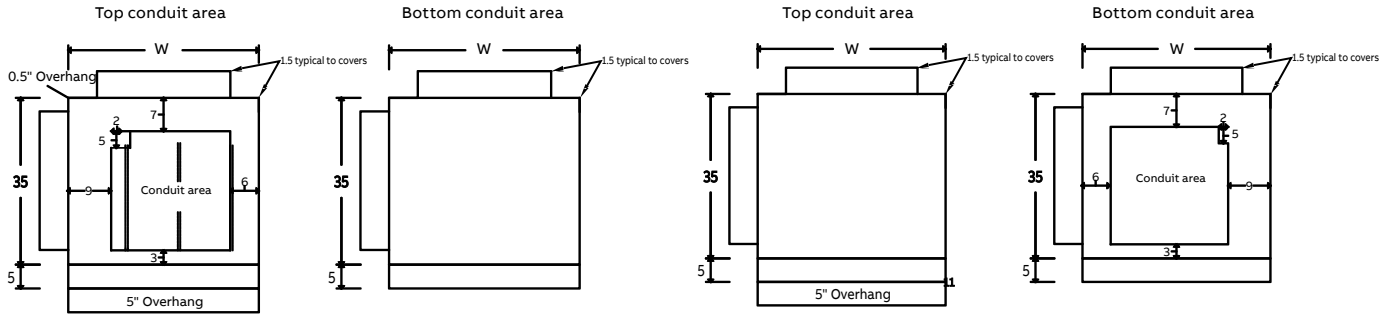
1200 A–3000 A: W= 40"
4000 A: W= 45"
NEMA Type 1 height: 90"

Conduit area

NEMA Type 3R enclosures

Main lug section — MLO

Figure 1A

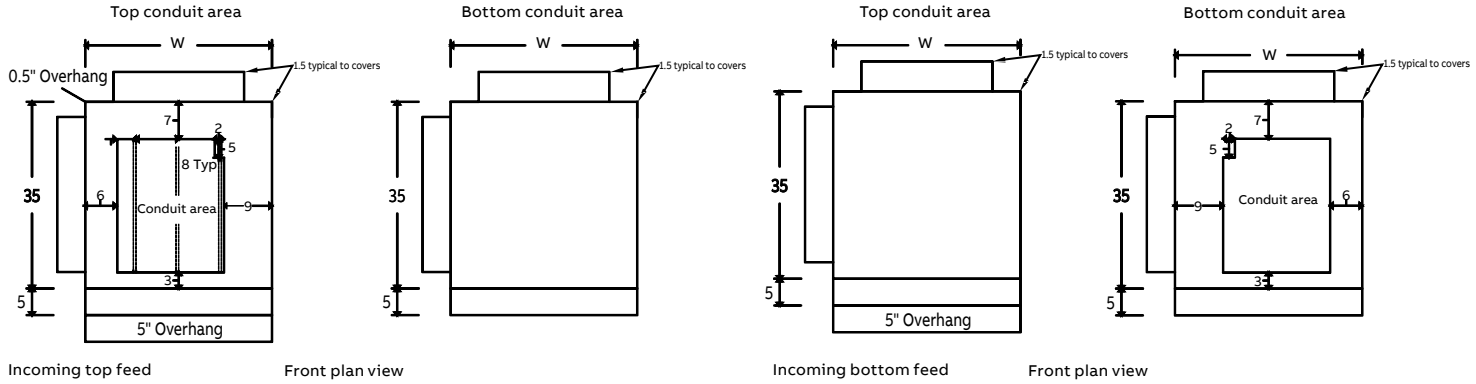


Incoming top feed Front plan view Incoming bottom feed Front plan view

1200 A–3000 A: W= 35"
4000 A: W= 40"
NEMA Type 3R height: 91"

Bussed pull section

Figure 2A



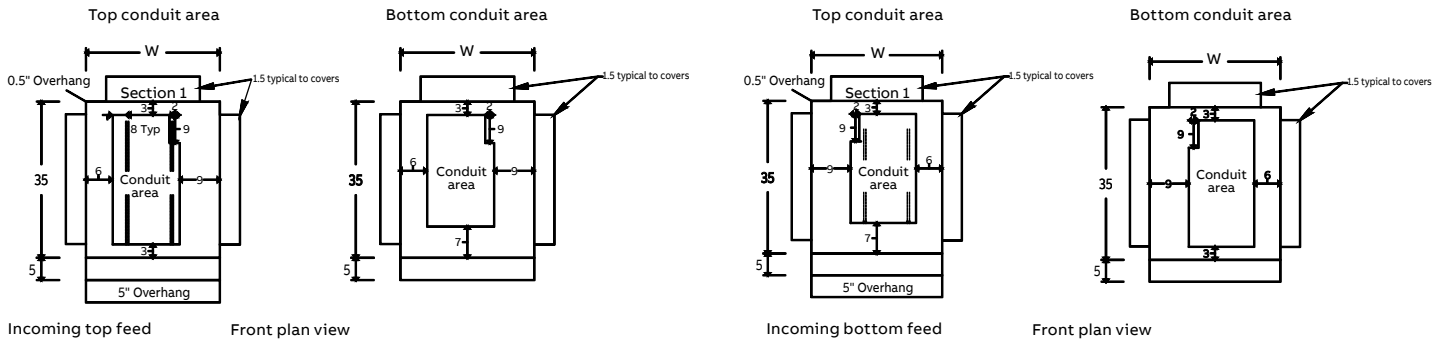
Incoming top feed Front plan view Incoming bottom feed Front plan view

1200 A–3000 A: W= 35"
4000 A: W= 40"
NEMA Type 3R height: 91"

—

Main section — main disconnect (ONLY section, fully rated without bus extension)

Figure 3A

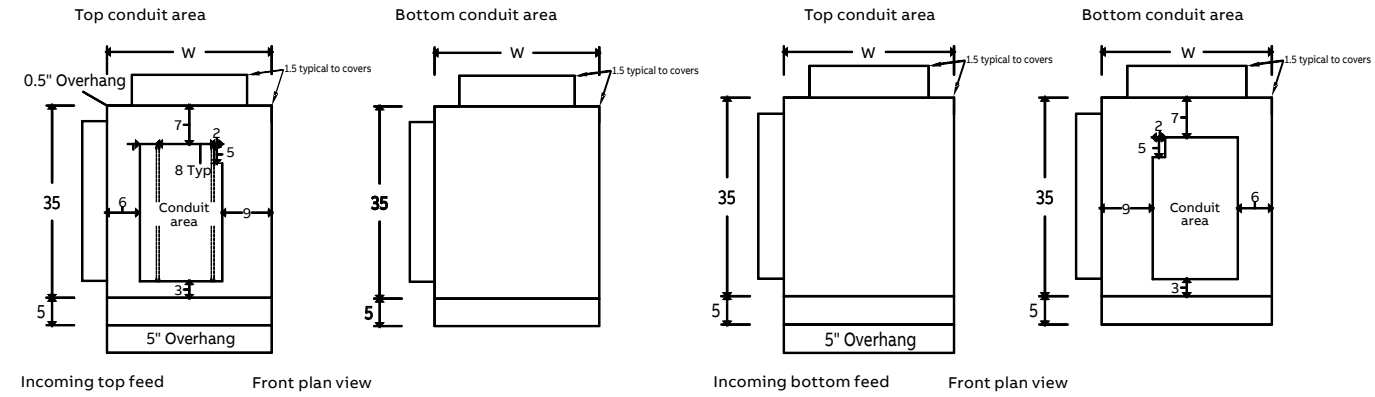


1200 A –3000 A: W= 30"
4000 A: W= 40"
NEMA Type 3R height: 91"

—

Main section — main disconnect (fully rated with bus extension)

Figure 4A



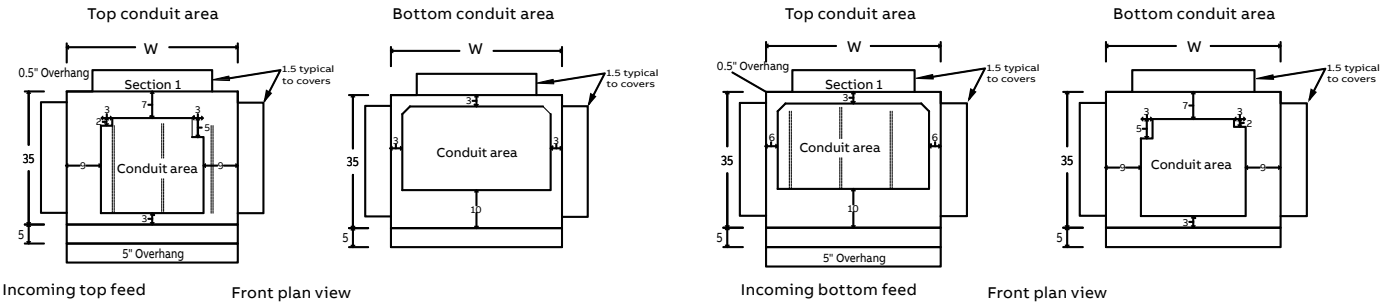
1200 A–3000 A: W= 30"
4000 A: W= 40"
NEMA Type 3R height: 91"

Conduit area

NEMA Type 3R enclosures

Group-mounted main sections (1200 A XT7) with feeder distribution

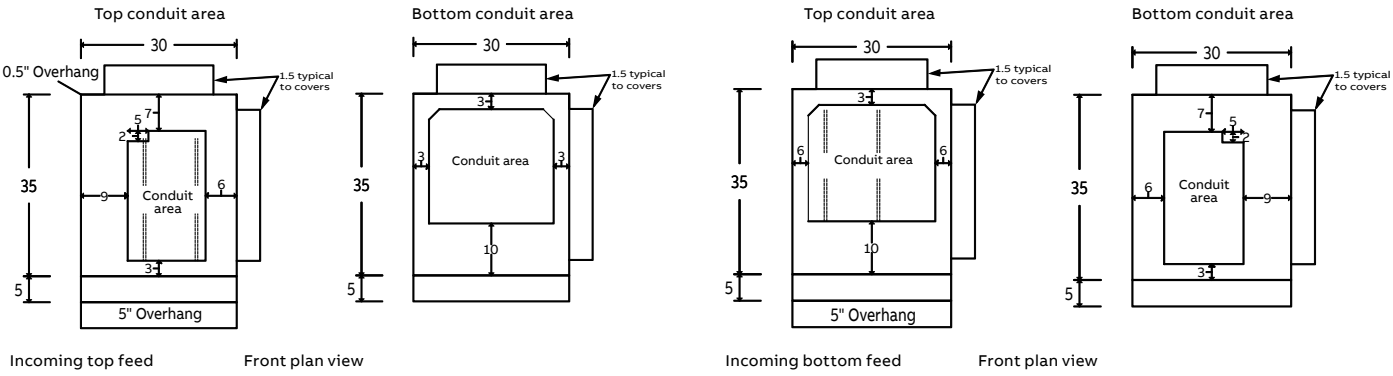
Figure 5A



One-sided medium: W= 40"
One-sided large: W= 45"
Offset extra-large: W= 50"
NEMA Type 3R height: 91"

Feeder distribution section (group-mounted), small, one-sided 30" (1200 A–2000 A)

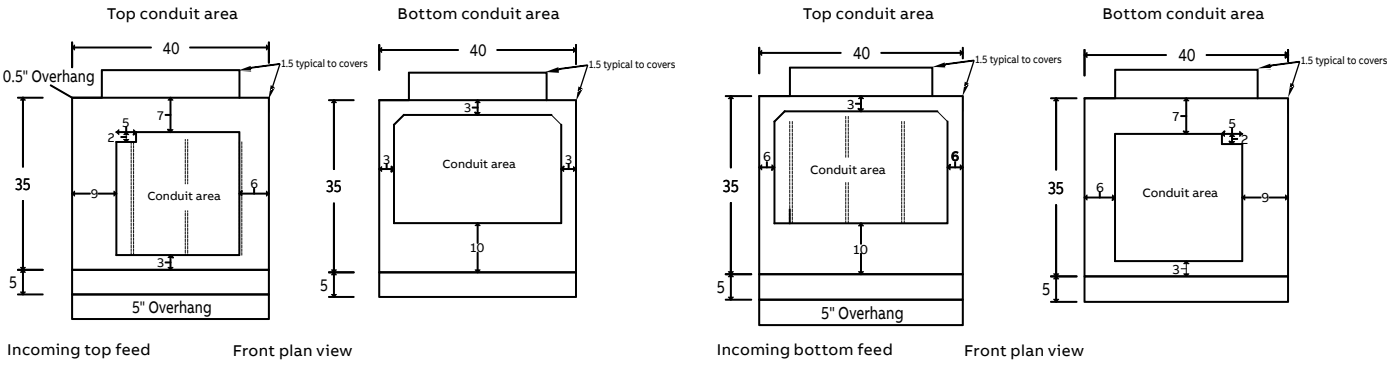
Figure 6A



NEMA Type 3R height: 91"

Feeder distribution section (group-mounted), medium, one-sided 40"

Figure 7A

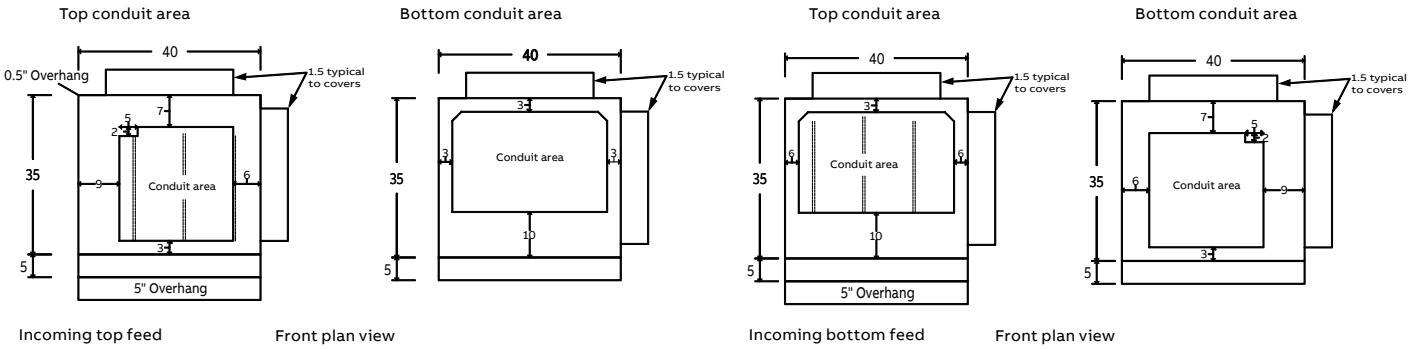


NEMA Type 3R height: 91"

—

Feeder distribution section (group-mounted), medium, offset 40"

Figure 8A



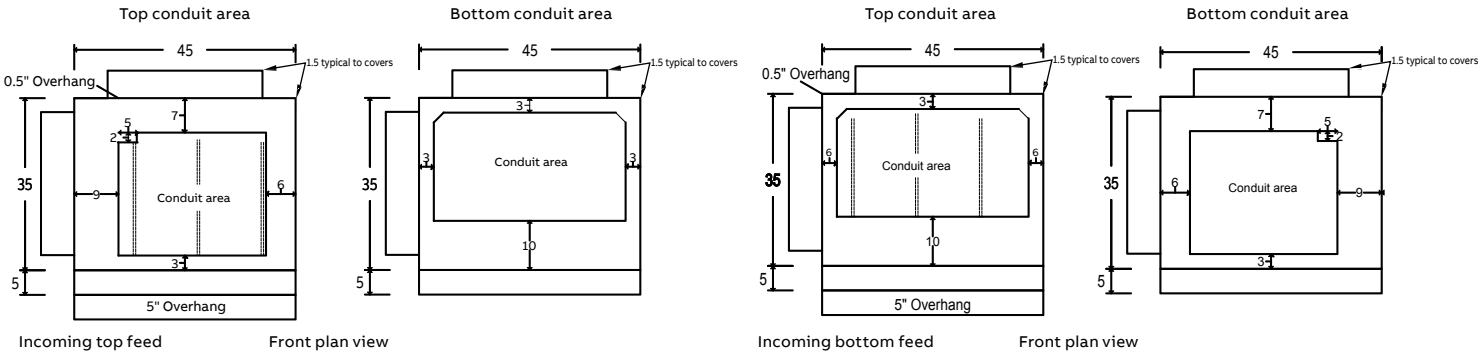
Incoming top feed Front plan view Incoming bottom feed Front plan view

NEMA Type 3R height: 91"

—

Feeder distribution section (group-mounted), large, one-sided 45"

Figure 9A



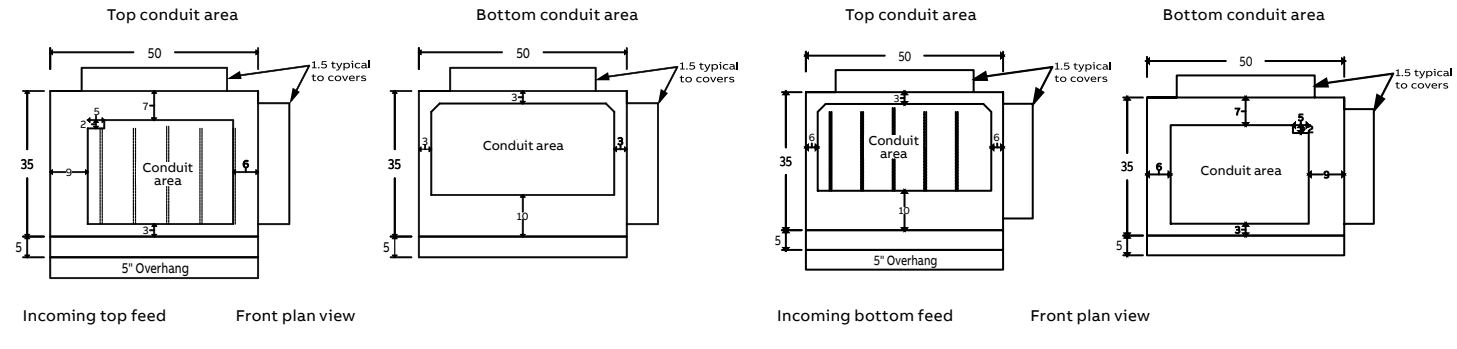
Incoming top feed Front plan view Incoming bottom feed Front plan view

NEMA Type 3R height: 91"

—

Feeder distribution section (group-mounted), extra-large, offset 50"

Figure 10A



Incoming top feed Front plan view Incoming bottom feed Front plan view

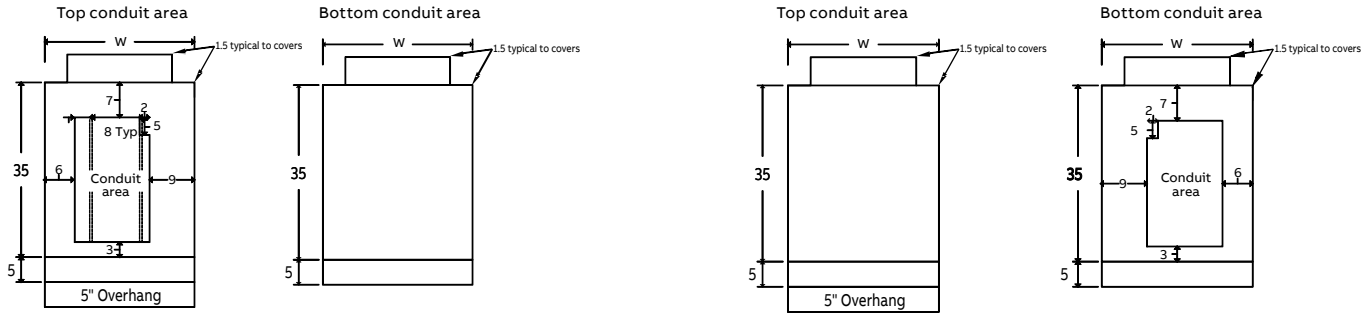
NEMA Type 3R height: 91"

Conduit area

NEMA Type 3R enclosures

Feeder distribution section (individually mounted)

Figure 11A

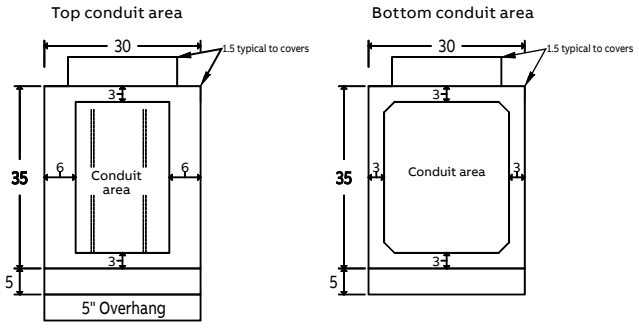


Incoming top feed — Front plan view — Incoming bottom feed — Front plan view

1200 A–3000 A: W = 30"
4000 A: W = 40"
NEMA Type 3R height: 91"

Blank pull section 30"

Figure 12A



Incoming top feed
Outgoing top exit

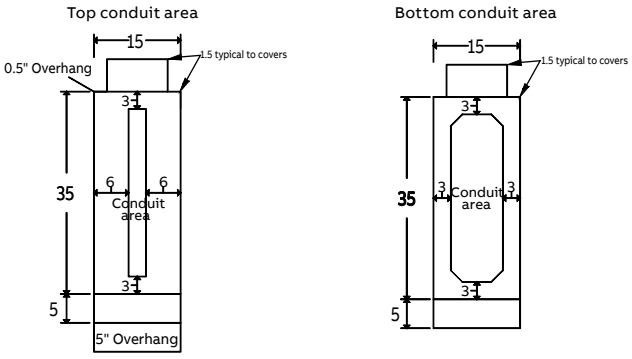
Front plan view

Incoming bottom feed
Outgoing bottom exit

NEMA Type 3R height: 91"

Blank pull section 15"

Figure 13A

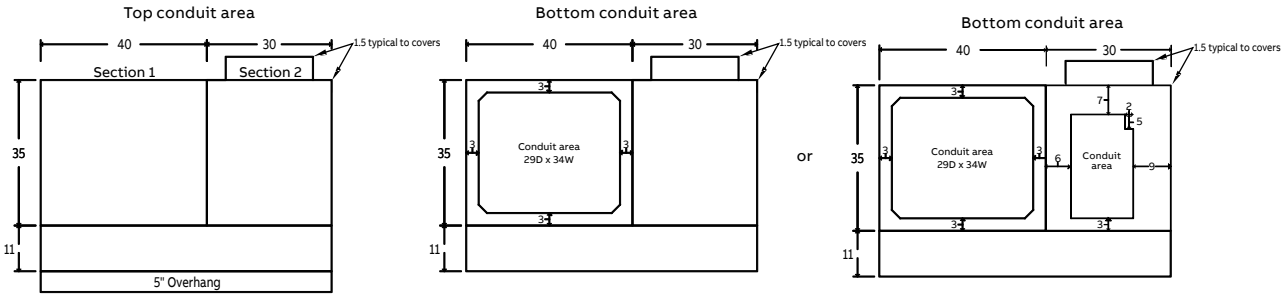


Incoming top feed — Front plan view — Incoming bottom feed

NEMA Type 3R height: 91"

EUSERC utility sections 1200 A–2000 A

Figure 14A



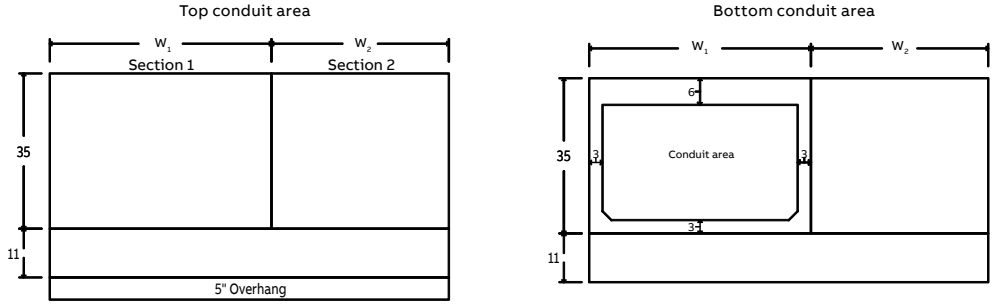
Incoming bottom feed

Front plan view

NEMA Type 3R height: 91"

EUSERC utility section with built-in main and separate EUSERC pull section

Figure 15A



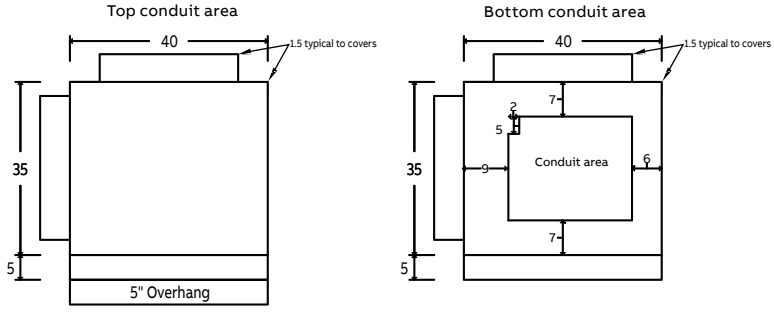
Incoming bottom feed

Front plan view

NEMA Type 3R height: 91"
1200 A–2000 A: W_1 = 40" and W_2 = 40"
2500 A–3000 A: W_1 = 50" and W_2 = 40"
4000 A: W_1 = 50" and W_2 = 50"

Other utility hot sequence utility section 1200 a–3000 a

Figure 16A



Incoming bottom feed

Front plan view

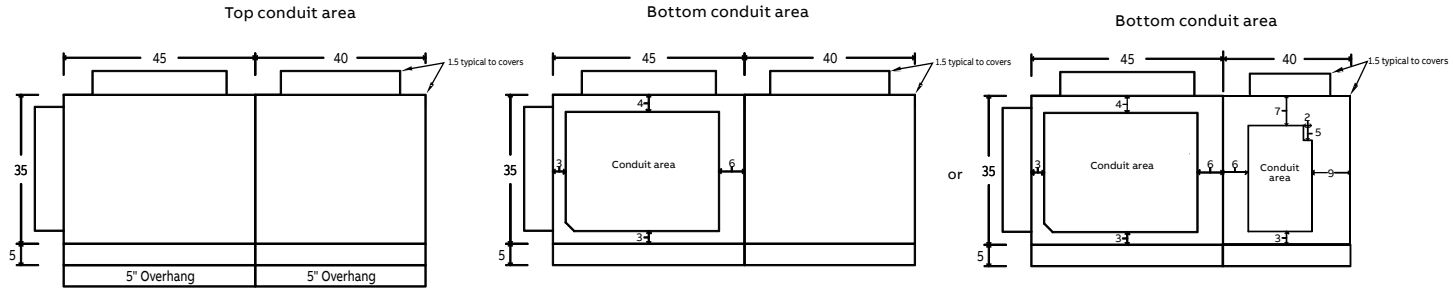
NEMA Type 3R height: 91"

Conduit area

NEMA Type 3R enclosures

Other utility hot sequence utility section 4000 A

Figure 17A



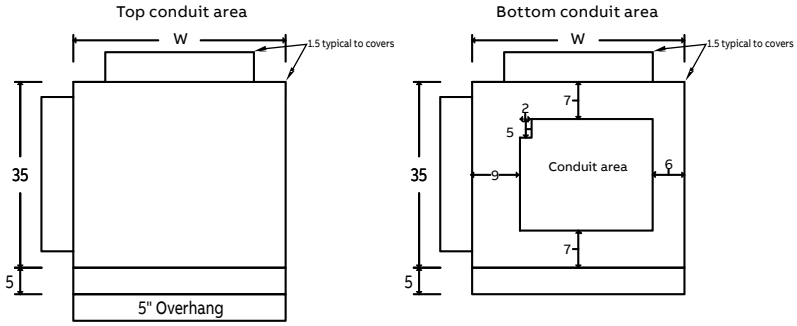
Incoming bottom feed

Front plan view

NEMA Type 3R height: 91"

Other utility cold sequence utility section

Figure 18A



Incoming bottom feed

Front plan view

1200 A–3000 A: W= 40"
4000 A: W= 45"
NEMA Type 3R height: 91"

REFERENCE STANDARDS

All ABB switchboards meet the following standards:

- UL 50 Thirteenth Edition — Enclosures for Electrical Equipment, Non-Environmental Considerations
- UL 50E Third Edition — Enclosures for Electrical Equipment, Environmental Considerations
- UL 67 Thirteenth Edition — Standard for Panelboards
- UL 98 Fourteenth Edition — Enclosed and Deadfront Switches
- UL 489 Fourteenth Edition — Molded-Case Circuit Breakers, Molded-Case Switches and Circuit Breaker Enclosures
- UL 512 Fuse Holders
- UL 891 Twelfth Edition — Switchboards
- UL 969 Sixth Edition — Standard for Marking and Labeling Systems
- UL 1066 Fifth Edition — Standard for Low-Voltage AC and DC Power Circuit Breakers Used in Enclosures
- National Electric Code 2023 Edition
- NEMA AB-1 Circuit Breakers, Molded Case and Circuit Breaker Enclosures
- NEMA PB1 2011 Edition — Panelboards
- NEMA PB2 2011 Edition — Deadfront Distribution Switchboards
- CSA C22.2 No. 244:19 2024 Edition — Switchboards
- Federal Specifications Panelboards W-P-115A, Molded Case Circuit Breakers WC-357B/GEN, Fusible Switches W-S-865C
- EPD (Environmental Product Declaration) 1SQC088156D0201
- EUSERC Standard — EUSERC 2023 Manual (Electric Utility Service Equipment) Seismic Certification
- Seismic certification
  - CBC 2022 — California Building Code
  - IBC 2021 — International Building Code
  - OSHPD
  - Levels 2.5g SDS (z/h:0) and 2.0g SDS (z/h:1)

INSTALLATION MANUALS

ADDITIONAL TECHNICAL INFORMATION, INSTRUCTIONS AND INSTALLATION GUIDELINES CAN BE FOUND IN THE FOLLOWING DOCUMENTS:

ABB Library		
Drawing selector tool		
1SQC900005M0201	ReliaGear® neXT	Reduced energy let-through (RELT) unit
1SQC900006M0201	ReliaGear neXT	Surge protective device (SPD) unit — group mounted
1SQC900006M0201	ReliaGear SB	Surge protective device (SPD) unit — individually mounted
1SQC900010M0201	ReliaGear neXT	Service entrance barrier
1SDH000719R0003	SACE® Tmax® XT	Installation instructions XT1
1SDH000721R0003	SACE Tmax XT	Installation instructions XT2
1SDH000722R0001	SACE Tmax XT	Installation instructions XT4
1SDH002011A1002	SACE Tmax XT	Installation instructions XT5
1SDH002012A1002	SACE Tmax XT	Installation instructions XT6
1SDH001821A1002	SACE Tmax XT7	User manual for use and maintenance of Ekip Touch Trip units for Tmax XT7 low voltage molded-case circuit breakers
1SDH000719R0610	SACE Tmax XT	PLL XT1–XT4 padlock device OP/CL
1SDA066592R1	SACE Tmax XT	PLL XT2–XT4 padlock device OP/CL
1SDH002013A1604	SACE Tmax XT	PLL padlock device XT7 in open position
1SXU210248C0201	SACE Tmax XT UL/CSA	Low voltage molded case circuit breakers UL 489 and CSA CS22.2 standards for the NEMA market
1SDH001000R0524	Ekip Signaling 2K	E1.2-E2.2-E4.2-E6.2-XT2-XT4-XT5-XT7-XT7M
1SDC210199D0206	Tmax XT UL	Technical characteristics 4.2021
1SDC210199D0206	Tmax XT UL/CSA	Technical characteristics LV MCCB
11SDC210066D0201		Guide to instantaneous selectivity
DET-008/1SQC930152D0228		Series rating
DET 760		Selectivity coordination
1SQC088002C0201	ReliaGear SB	Technical application guide
1VAL088301-MB	ReliaGear SB	Installation, operations and maintenance manual
1SDH002311A1002	SACE Emax 3	Operating instructions

DEFINITION OF TERMS

Accessible means not permanently closed in by the switchboard section structure, and capable of being inspected and maintained through access plates or doors without disturbing the switchboard section structure.

Ambient temperature is the temperature of the surrounding air that meets a fuse, circuit breaker or conducting bus within a switchboard.

Amp rating is the amount of current a fuse will carry continuously without deterioration, or a circuit breaker without tripping, and without exceeding temperature rise limits specified for a particular fuse or circuit breaker by NEC requirements and UL standards.

Amp setting — adjustable varies the continuous current-carrying ability of a circuit breaker through a predetermined range.

Ampacity is the current-carrying capacity in amps a conductor can carry continuously under the conditions of use without exceeding its temperature rating.

Available short-circuit current is the maximum rms (root-mean-square) symmetrical current at a given point in a power system, operating with maximum generating capacity and connected load, and can deliver to any zero-impedance short circuit applied at that given point.

Branch circuit is the circuit conductor between the final overcurrent device protection and the outlets or point of use.

Enclosure is a constructed case to protect personnel against contact with the enclosed equipment and to protect the enclosed equipment against environmental conditions.

Frequency is the number of complete cycles of AC voltage that occur in one second, and is measured in hertz.

Group mounted device is one of a closely grouped assembly of devices that is mounted on a common base or mounting surface using panelboard-type construction. The total assembly is then mounted in a switchboard intermix or distribution section.

In addition to feeder circuit breakers, main circuit breakers can be group-mounted for applications of 1200 A and below. Group-mounted circuit breakers are typically standard rated for up to 80 percent of the rated amperage. 100 percent rated circuit breakers are available in Tmax® XT7 and Emax 3 circuit breakers, with some limitations.

Individually mounted device is one that is not panel mounted, and that may or may not be enclosed in its own section. Separated from other devices to minimize undesired influence from them.

Insulation level — Switchboards have an insulation level rating at twice the rated voltage plus 1000 volts. The dielectric test for this rating consists of applying a 60 hertz voltage for one (1) minute phase-to-phase and phase-to ground with switching devices closed.

Line refers to the incoming (live) side of equipment or device.

Load is the outgoing (switched) side of equipment or device.

Definition of terms

Lug is a metal connector or terminal that is used to attach electrical conductors to a device or component. In this context, the reference is to lugs installed on the device or component to connect cables, not to lugs that have been crimped onto the cables themselves.

- **Main lug** is the connecting means between the incoming service cable and the main bus bar.
- **Mechanical lug** is a terminal with one or more wire binding screws that are tightened to hold the conductor or cable.
- **Compression lug**, also called a crimp lug, is a lug that is crimped to hold cable.

Main device is a single device that disconnects all ungrounded switchboard conductors, other than control power conductors when used, from the supply bus. Note that line conductors are not switched.

NEMA Type 1 enclosure — general purpose indoor is intended primarily to prevent accidental contact of personnel with the enclosed equipment. The enclosure is intended for use indoors where it isn't exposed to unusual service conditions, dripping or splashing water, steam or conductive dusts (NEMA PB2-5.02).

NEMA Type 3R outdoor enclosures are intended for use in wet locations or outdoors to protect the enclosed equipment against rain. They are sleet (ice) resistant but are not dust, snow or sleet (ice) proof. If conduit connections are provided, these enclosures shall have a conduit hub or equivalent provision for watertight connections at the conduit entrance when the conduit enters at a level higher than the lowest live part. They have provisions for drainage.

Selective tripping is the application of circuit breakers or fuses in series, so that, of the circuit breakers or fuses carrying fault current, only the one nearest the fault opens and isolates the faulted circuit from the system.

Series-connected rated panel means the UL listed short-circuit rating of the panel is equal to the IC rating of the main protective device when properly applied with branch circuit protective device.

Service disconnect is a means to disconnect all ungrounded conductors in a building or other structure from service conductors.

Service entrance equipment usually consists of circuit breakers or switches and fuses and their accessories. It is located near the point of entrance of supply conductors to a building and is intended to constitute the main control and means of cutoff for the supply to the building. Service entrance equipment has a removable link between the neutral bus and ground bus. The main device is barriered from feeder devices.

Shunt trip opens a circuit protection device by remote control.

Space and busing for future. When space and busing is only required for future inclusion of any specified devices, appropriate device supports, proper bus connections and corresponding vertical bus are furnished.

Switchboard per the NEC is a large single panel, frame or assembly of panels on which the following can be mounted on the face, the back or both: switches, overcurrent and other protective devices, including buses and instruments. Switchboards are generally accessible from the rear as well as from the front and are not intended to be installed in cabinets.

Switchboard short-circuit ratings. Switchboards shall be applied on a system having an available short-circuit current not greater than the short- circuit rating of the switchboard. All switchboards are marked with a short-circuit rating. The switchboard will be fully rated, or series-connected rated. When fully rated, the short-circuit rating of the switchboard is that of the lowest rated device in the switchboard. Series-connected ratings are based on short-circuit ratings given to two or more devices connected in series. In either case, the bus bar bracing must be equal to or exceed the short-circuit rating of the switchboard. Bus bar bracings available are 65 kA, 100 kA, 150 kA and 200 kA rms symmetrical amperes; 65 kA is the standard bracing.

Utility compartment is a compartment within a switchboard section meeting utility standards designed to accommodate current and potential transformers provided by the utility company.

- Hot sequence utility compartment is mounted on the line side of the main disconnect.
- Cold sequence utility compartment is mounted on the load side of the main disconnect.

Voltage is electrical pressure that moves electrons through a conductor and is measured in volts.

Voltage rating is the rms (root-mean-square) of alternating current voltage at which a fuse or circuit breaker is designed to operate.

Working space. The National Electrical Code (NEC) Article 110.26 requires that sufficient access and working space shall be provided and maintained around electrical equipment to permit ready and safe operation and maintenance of such equipment. The minimum working space required by the NEC for switchboards 150 volts-to-ground is three (3) feet. For voltages over 150 volts-to-ground, three feet is required from the enclosure to insulated surfaces, three and one-half (3½) feet to grounded surfaces, including concrete, and four (4) feet to exposed live parts. Working space is required from all accessible surfaces (front, rear, sides). Working space is not required in back of switchboards (Class 1, Class 2) when all connections are accessible from locations other than the back.

X value is an arbitrary vertical measurement of the usable mounting space on a panelboard and/or switchboard for a fusible switch, circuit breaker or accessory. X is equal to 1.385" or 35.2 mm. Height of the interior is the sum of the horizontal panel-mounted components or spaces.

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